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1 Motivation

In this section we want to give an overview of the topic we are focusing on as well as our motivation to approach this topic, answering the question of, why you should care about our idea.

Attention Deficit Hyperactivity Disorder - better known as ADHD - affects millions of people worldwide. It is a highly diverse neurodevelopmental condition that brings unique challenges, but also hidden strengths. Experiencing life with ADHD means thinking, feeling, and acting in ways that often differ from the norm.

One of our team members is currently undergoing diagnosis for ADHD, which enables us to empathise deeply with its characteristics and challenges. Hence, our motivation is not only societal but also personal: we wish to develop focus- and productivity-enhancing solutions tailored to the specific needs of affected individuals while raising awareness and thus, reducing stigma. We believe in a holistic approach to create a support system that goes beyond medication. Taking into account unique strengths of neurodiverse minds and experiences from affected individuals, we seek to identify suitable methods, tools and habit building strategies.

Raising Awareness: ADHD is among the most often diagnosed mental conditions, affecting about 5% of adults worldwide (Polanczyk et al., 2015). Nonetheless, many individuals - especially women - remain undiagnosed (Babinski and Waschbusch, 2016). Typical ADHD symptoms include impulsivity, attention deficit, hyperactivity and affective lability which typically result in problems with task and time management (e.g., always being late), chronic procrastination, forgetfulness, and mental overload. More severe consequences include a 1.5 times higher risk of developing substance dependencies (Lee et al., 2011) and a more than twofold increased risk of depression (Zhang et al., 2025). Despite these risks, ADHD continues to be misunderstood and trivialized – often reduced to the cliché of the inattentive young boy. Raising awareness is crucial not only for early diagnoses and intervention, but also to help affected individuals develop personalised coping mechanisms and build healthier habits from early on.

Supporting Neurodiverse Minds: In our initial round of interviews with ADHD students, we found that many already rely on various productivity tools such as calendars, reminders, to-do lists and notebooks. However, maintaining these tools regularly and with required consistency is often becoming a challenge already. Interviewees reported difficulties with prioritisation, breaking down complex tasks, and estimating task durations realistically, yet our findings quickly went beyond standard task management issues. Core challenges shared by Interviewees included:

- Initiation paralysis
- "All-or-nothing"-mindset
- Frequent task-switching and short attention spans
- Keeping up with tight schedules
- Needing just the right amount of pressure to perform, else procrastinating with too little or being overwhelmed by too much pressure
- Quick loss of motivation
- Mental overload and high distractability - whether by internal thoughts or external stimuli like social media
- Poor sense of time
- Restlessness and excess physical energy

These insights shifted our focus: rather than building a standard productivity tool, we aim to create an adaptive support system that understands and embraces ADHD-related needs. Our goal is to work with, not against, neurodiverse minds.

Implementing Positive Changes: Getting out of bed and completing a small, feasible task right away creates not only the first sense of achievement but may boost motivation for that day. Several interviewees confirmed that a productive morning is an essential driver for setting both mood and productivity for the following hours. Further, some even stated that the opposite would ruin the feeling of productivity for the entire day, echoing the “all-or-nothing”-mindset mentioned earlier. This inspired our aim to support users in developing and maintaining productivity-increasing routines without becoming overwhelming or adding too much pressure. Simplicity emerged as a core requirement from interviews: users wished for tools that “keep it simple,” reduce friction, allow flexibility, and avoid rigid structures that feel overly restrictive. Despite shared symptoms, our interviews underlined each person’s individual needs and strategies to increase productivity, highlighting the importance of allowing flexibility within our tool. For some affected individuals, allowing to switch between topics and thus enabling parallel working on different tasks was beneficial for their productivity, whereas some could only focus on one thing at a time or else it would result in chaos. This important discovery shaped our mission of building an adaptive system that personalises support based on individual behaviours, preferences, and contexts.

Connecting Like-Minded Individuals: A simple, but probably the most powerful insight came from our team member’s experience about attending a weekly student workshop for those struggling with ADHD symptoms, whether being officially diagnosed yet or not. It was neither expert advice nor generic tips that made the workshop special, but the connection with other like-minded people – sharing strategies, voicing frustrations, hearing others feel or experience the exact same, feeling truly connected with more than one

person at the same time, never being the only one that is too late, exceeding breaks because all get caught up in conversation whilst smoking outside, switching topics without losing word, being okay with interrupting each other because everybody understands how it feels like to keep the thought (impossible), but most importantly: feeling collectively understood. Without any exception, every interviewee agreed on how easily and effortlessly people with ADHD generally connect with each other. This sparked one of our main feature ideas *NeuroConnect*: a platform accessible for anyone with ADHD symptoms. We aim to bring together like-minded individuals locally and online, to exchange ADHD-related experiences, offer support, exchange strategies, and potentially build lasting routines - or even friendships. But first and foremost, *Neuro-Connect* shall be designed to build a community for everyone to understand, accept and love their neurodiverse minds.

As part of our design thinking process, we conducted a comparative analysis of existing apps targeting ADHD management, productivity, and focus enhancement. We analysed 13 relevant applications, ranging from task planners (e.g., Todoist, Me+) and productivity timers (Forest, Brain Focus) to more specialised ADHD tools (Inflow, Univi, Splitti). Among these, Splitti emerged as the most comprehensive and closely related to our concept. It combines AI-assisted task breakdown, intelligent prioritisation (Pomodoro, Eisenhower), and user profiling with motivational nudges, mood-friendly audio, and community features. Discovering an app that closely mirrors several of our own ideas was initially a surprising moment for our team. However, this finding has ultimately strengthened our motivation: Instead of viewing Splitti as a barrier to innovation, we see it as a confirmation that our problem space is real, pressing, and already being explored. Hence, our solution is not only relevant but potentially market-viable. At the same time, we identified several ways in which our app concept meaningfully differentiates itself:

- Deeper Personalisation: While Splitti provides general organisational support, our app focuses on the intelligent connection between unstructured thoughts and structured task planning, using an integrated AI assistant that can derive tasks from brain dumps and adaptively surface only the most essential next step.
- ADHD-Specific Forums: Our discussion spaces are structured along the three main ADHD symptom categories - impulsivity, hyperactivity, and attention-deficit, allowing for more targeted peer support and strategy sharing. We also have another category for diagnosis, as we recognise that especially individuals who are newly diagnosed or suspect they might have ADHD often face a confusing and overwhelming journey.
- Focus + Social Synergy: We believe our strongest innovation lies in the integration of solo productivity and real-life social interaction. With the Connect screen, users can share achievements (in a way inspired by platforms like Strava), join physical co-working meetups, and motivate each other through live focus sessions.
- Method Library with Adaptive Logic: Unlike static collections of techniques, our Browse section offers a curated yet dynamically tailored set of ADHD-specific and general productivity strategies, based on user behaviour, preferences, and feedback.

Encountering similar apps has motivated our group more to build a lovable and useful product. It shows us that we are on the right track with our idea. We also know from our interview that these solutions are needed, this reinforced this sentiment and allows us to analyse the less lovable aspects of the existing solutions and to bridge identified gaps and refine our concept to add unique value. This has sharpened our design thinking approach and renewed our commitment to prototyping an app that is not just helpful, but also socially meaningful, user-centered, and grounded in real-world needs.

2 Overview of Our Project

In the following section we want to provide a short description and overview of our project, the main problems we identified and the core solutions we want to implement.

The overall aim of our application is to support individuals with ADHD in navigating everyday challenges more effectively. While we acknowledge that ADHD is a spectrum and experiences vary widely, our approach is grounded in insights gained through interviews with people affected by the condition. These conversations revealed recurring struggles that we believe are shared by many, and which we aim to address in a way that benefits as broad a group as possible. More specifically, our application allows users to externalise their thoughts through a **brain dump** function, stay focused when working on tasks and manage their time and tasks through the **focus** mode, connect with others who have ADHD through a community space called **connect**, and learn more about ADHD and help with providing coping strategies via a method catalogue, a feature we call **browse**. Together, these four elements help users to feel more confident with ADHD and about their ADHD related problems, manage their daily lives more easily, overcome their procrastination and become more productive, and motivate and support them instead of feeling overwhelmed and left alone with their problems and entangled in procrastination.

We identified three key areas of difficulty: first, the initial overwhelming challenges for newly diagnosed individuals, stemming from a minimal understanding of ADHD and how to manage it. Second, the ongoing struggles with managing tasks and time effectively, which directly relates to being productive. These types of problems can be attributed to how minds of people with ADHD function. This can also be seen in the way many are easily distracted from their thoughts while working on a task, or finding it difficult to start a task and tend to procrastinate. And finally, individuals with ADHD often struggle with effectively connecting and communicating with others, as they find it easier to interact with people who share similar experiences, leading to a greater sense of understanding and connection.

Since we are aware that ADHD manifests itself in many different ways and experiences vary considerably, our app is deliberately designed for a wide range of users. We aim to support those who are newly diagnosed and looking for orientation, connection, and better understanding of ADHD, as well as individuals who have lived with the diagnosis for years but struggle with traditional tools like to-do lists and calendars. By offering modular features, our app allows users to access the full range of functions while also addressing individual needs. Each feature targets a specific area of need, allowing users to engage only with the parts that are most relevant to them.

The "pain points" mentioned in the beginning of this section formed the foundation for the app's design. These issues will be addressed as follows, aiming to solve them or help users manage them more effectively. The brain dump feature offers a quick and effective way to record spontaneous thoughts. These thoughts often arise in the middle of other tasks and are so distracting that individuals with ADHD cannot continue working productively. Brain dumping provides a mentally freeing sensation, as the thought can be temporarily set aside, enabling users to maintain focus without the risk of forgetting valuable thoughts. These thoughts can be processed later and converted into a task for the focus section, a plan for the community space, or as a shared experience in the browse section. The focus mode enables users to organize and manage their tasks efficiently and stay concentrated while working on them. Users can define tasks, Neura helps to break them into smaller steps, which helps users get started working and overcome procrastination. The connect space allows users to view their friends' activity, intended to serve as a source of motivation. Further, users can plan co-working sessions that follow the body doubling principle, as well as plan other social activities to connect with others who experience ADHD. Lastly, the browse space serves as a method shopping catalogue where users can discover, try, and discuss ADHD-friendly techniques to find the methods that work best for them personally. Through these core features, our application aspires to provide both practical tools and a sense of community, helping users with ADHD feel more organised, empowered, and connected. By supporting users in their everyday life, which has a positive impact to their emotional well-being, our app makes a direct contribution to the **UN Sustainable Development Goal 3: Good Health and Well-Being**. We want to help people with neurological diversity manage their time better, be more productive, reduce stress, and feel more connected to others.

3 Design Thinking

In this Section we want to give a brief overview of Why we used Design Thinking.

We used design thinking because, even within our own team, we recognised that having or not having ADHD impacts how deeply one can empathise and immerse themselves in someone else's experience. So the question really is: Why should a team without prior connection or experience to a topic be able to build a lovable app? Most of our team didn't have a personal connection to ADHD. That's why, until now, we have interviewed seven people with ADHD. We quickly realised that ADHD is a spectrum. Even though all our interview partners come from the same demographic background, very different patterns and experiences emerged. That's exactly why design thinking is so important for us in this context. As emphasized by Brown et al. (2008), design thinking helps ensure we don't end up designing an app that solves a problem that doesn't actually exist.

3.1 User Persona

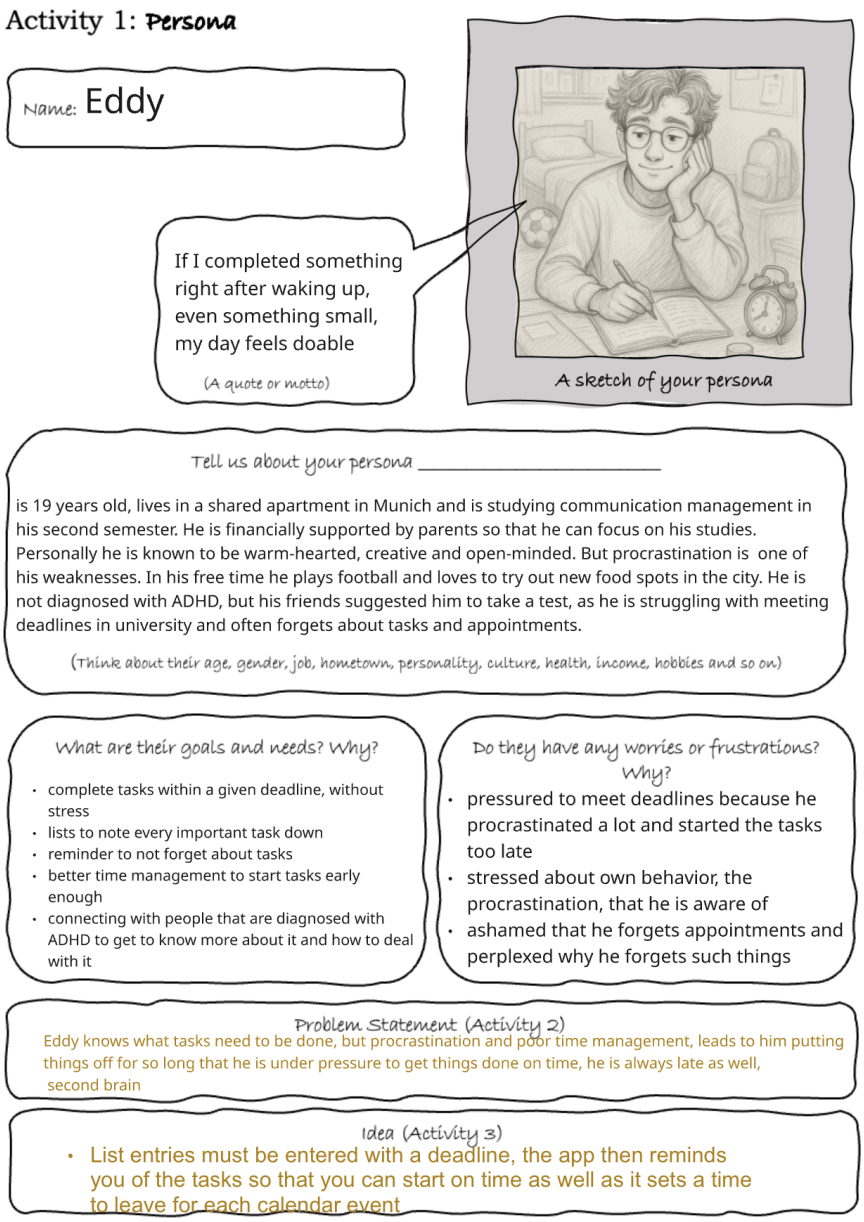


Figure 1: User Persona of Eddy

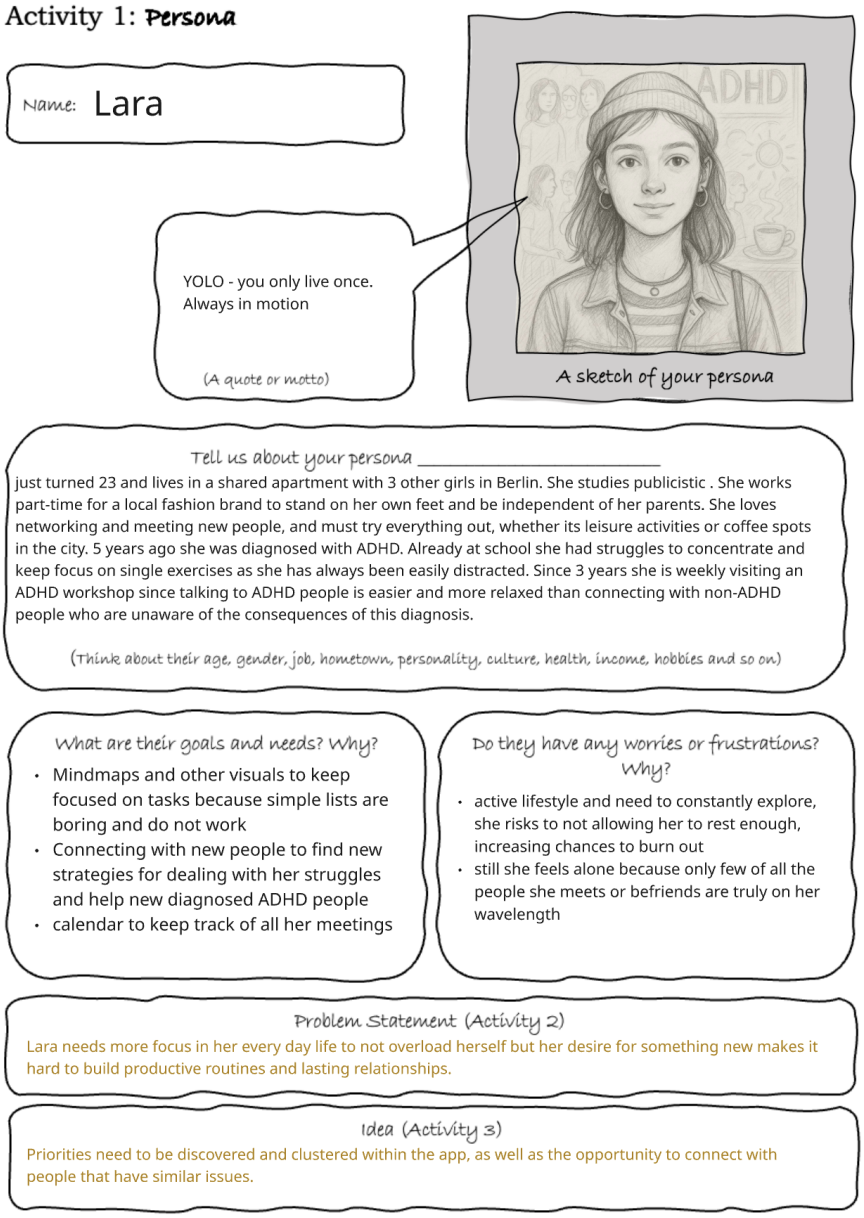


Figure 2: User Persona of Lara

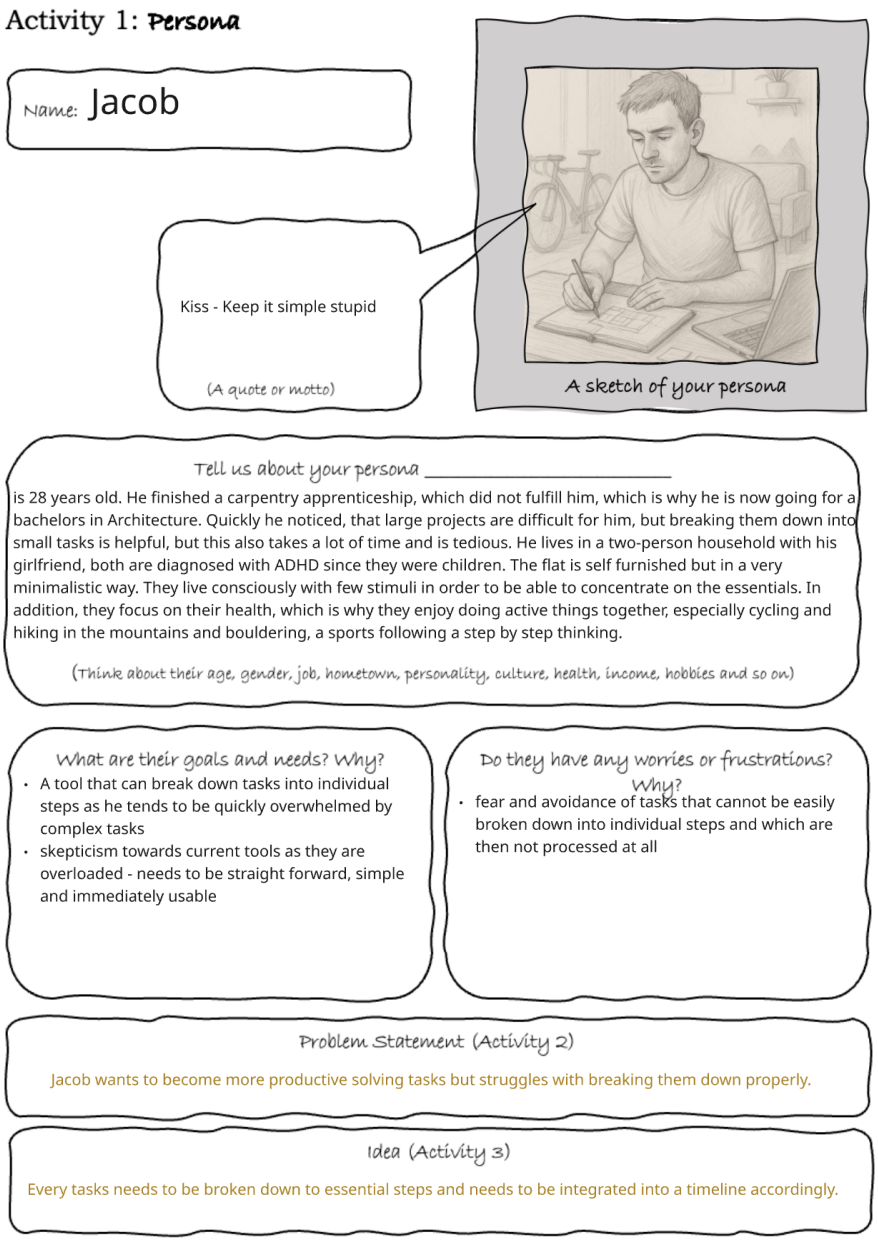


Figure 3: User Persona of Jacob

3.2 Define

Based on our user personas, we analysed the needs and problems of the exemplary user groups. Figure 4 illustrates our findings from the define phase. The individual points are sorted in descending order of importance. Our identified problems can be clustered into three core problem areas:

- **Time and Task Management**
- **Lack of Structure and Organisation**
- **Mental and Social challenges**

Time and Task Management The first identified core issue, having difficulties with time and task management, manifests itself particularly in the fact that ADHD sufferers find it difficult to organise tasks according to their importance or urgency and often feel that they do not know where to start. Additionally, a frequent challenge is the sudden appearance of distracting thoughts or ideas.

These distract users from their current task and make it difficult for them to maintain focus. As a result, but not only because of this, procrastination is a recurring problem. This is further accompanied by many feeling overwhelmed by complex assignments and being unsure where to begin. Another related problem is simply missing tasks or deadlines.

Lack of Structure and Organisation Another major issue that emerged is the lack of structure and organisation in users' daily life. The lack of structure often arises from missing routines and ends with days being chaotic or unproductive. A direct consequence of this is frequent lateness, which can negatively impact social relationships and cause frustration. To address both time management and organisational challenges, users frequently rely on multiple apps and platforms, such as calendars and to-do lists. As these apps have to be used independently of each other, maintaining and coordinating these apps so that they can be used effectively takes a lot of time, as many actions are duplicated. This increasingly leads users to stop using these tools altogether.

Mental and Social challenges Finally, we identified a set of problems that are more psychological or social in nature. The feeling of being overwhelmed mentioned above can develop quickly into a persistent feeling and cause mental distress. Users also face feelings of inadequacy and self-doubt as they find certain tasks more difficult than others, which is also linked to a shortened attention span. Conversely, the desire to compensate for these negative emotions and thoughts often leads users to overcommit and overload themselves, which can lead to a further spiral of self-doubt and the feeling of being overwhelmed. These results serve as a foundation for the subsequent ideation process.

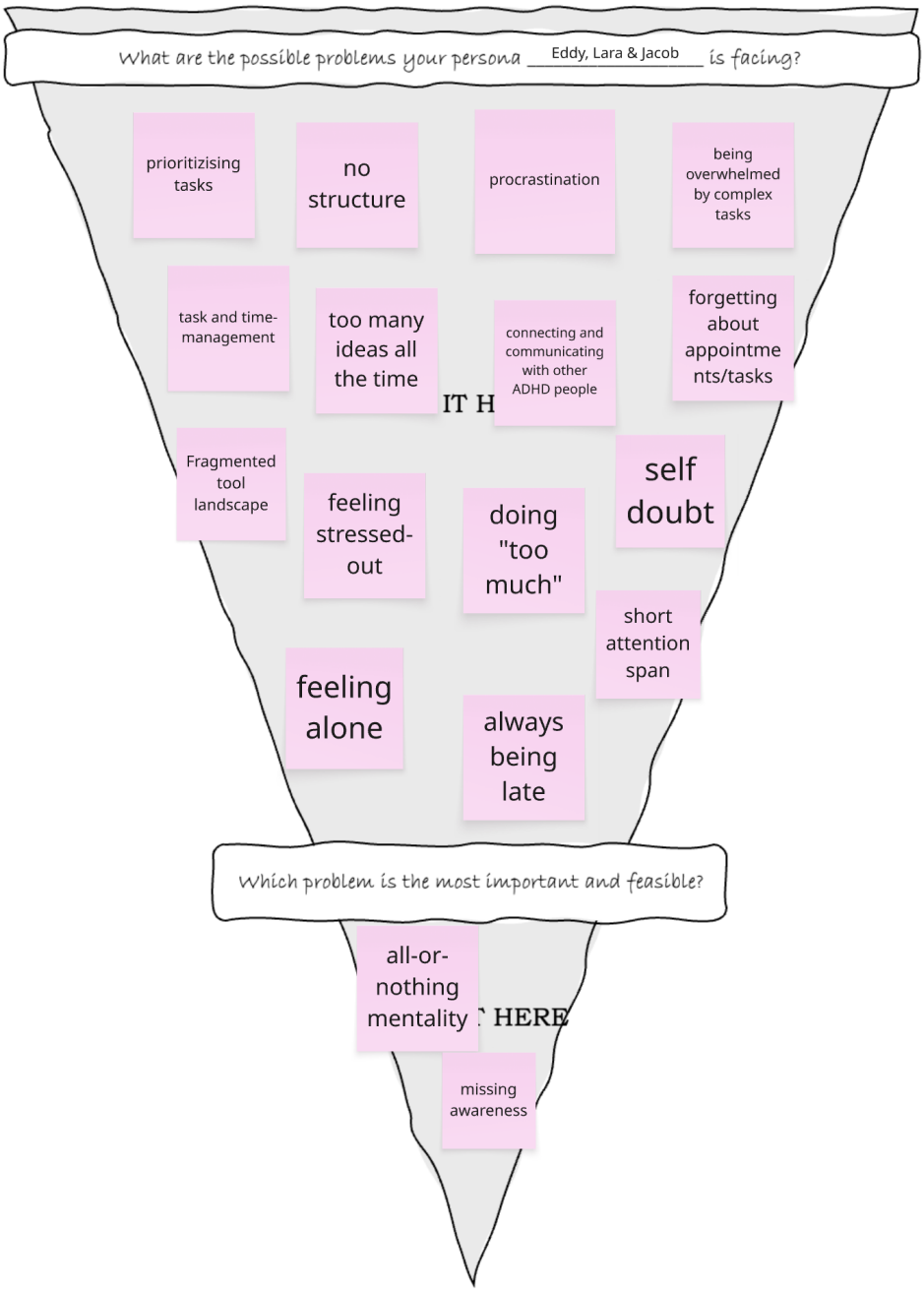


Figure 4: Activity of Problem Definition

3.3 Ideate

Building on the core problems identified we entered the ideation phase in which we developed a broad spectrum of innovative solution concepts. For a visual representation of these concepts, see Figure 5, where they're listed from most to least important. The following concepts are grouped according to the respective core challenge they address:

Time and Task Management To overcome difficulties with time perception, prioritisation and procrastination, we have brainstormed several solutions to support the structured handling of tasks:

- **Smart Task Assistant:** An intelligent tool that helps to break down complex tasks into small, manageable steps that prevent overwhelm and encourage the user to start with a complex task and work their way forward. Working through the small sub tasks should provide the user with a constant sense of achievement and gives the feeling of partial success.

- **Deadline and Reminder System:** Every task should be assigned a deadline. Based on this, the system should remind users of a task, considering both its deadline and urgency, to prevent missed deadlines and ensure an early start
- **AI-Based Time Planning:** The app learns from user behaviour, e.g. how long it takes to complete certain tasks or at what times the user has productive peaks, in order to make a customised, realistic daily and weekly schedule.
- **Focus Mode:** This function is designed to tackle the problem of being too easily distracted by blocking the use of apps such as social media so that the user stays focused on his actual task.

Lack of Structure and Organisation To reduce chaos and inefficiencies in a users daily life, the following ideas have come to our minds:

- **Second Brain:** This severs as an extended digital memory. It allows the user to dump every thought, note, ideate or similar kinds of information that is floating through the users head. This allows to put aside current unimportant thoughts and continue working productively at that moment. At a later point in time, the user can access the thoughts when they have time to process and organise them.
- **Topic based structuring:** The app should enable the user to display tasks and appointments based on different areas of life, e.g such as a differentiation between university stuff and leisure appointments.

Mental and Social Challenges To cope with psychological and emotional stress that ADHD users often face we have developed features that promote psychological well-being and connectedness to like-minded people:

- **Social Community:** The app should contain a social feature through which user facing ADHD problems can connect with others who have similar problems and interests
- **Goal Sharing:** To inspire and motivate each other to complete tasks, users can set and share their goals with friends
- **ADHD Education:** This informational feature is especially designed for users who are dealing with ADHD for the first time, but not exclusively, as people who have been dealing with ADHD for some time can still learn a lot. In this section users can learn a lot about ADHD symptoms but also coping strategies are presented, i.e how to deal with certain problems.
- **Guided Mediation:** These are short exercises that help users to reduce stress and improve focus in that moment

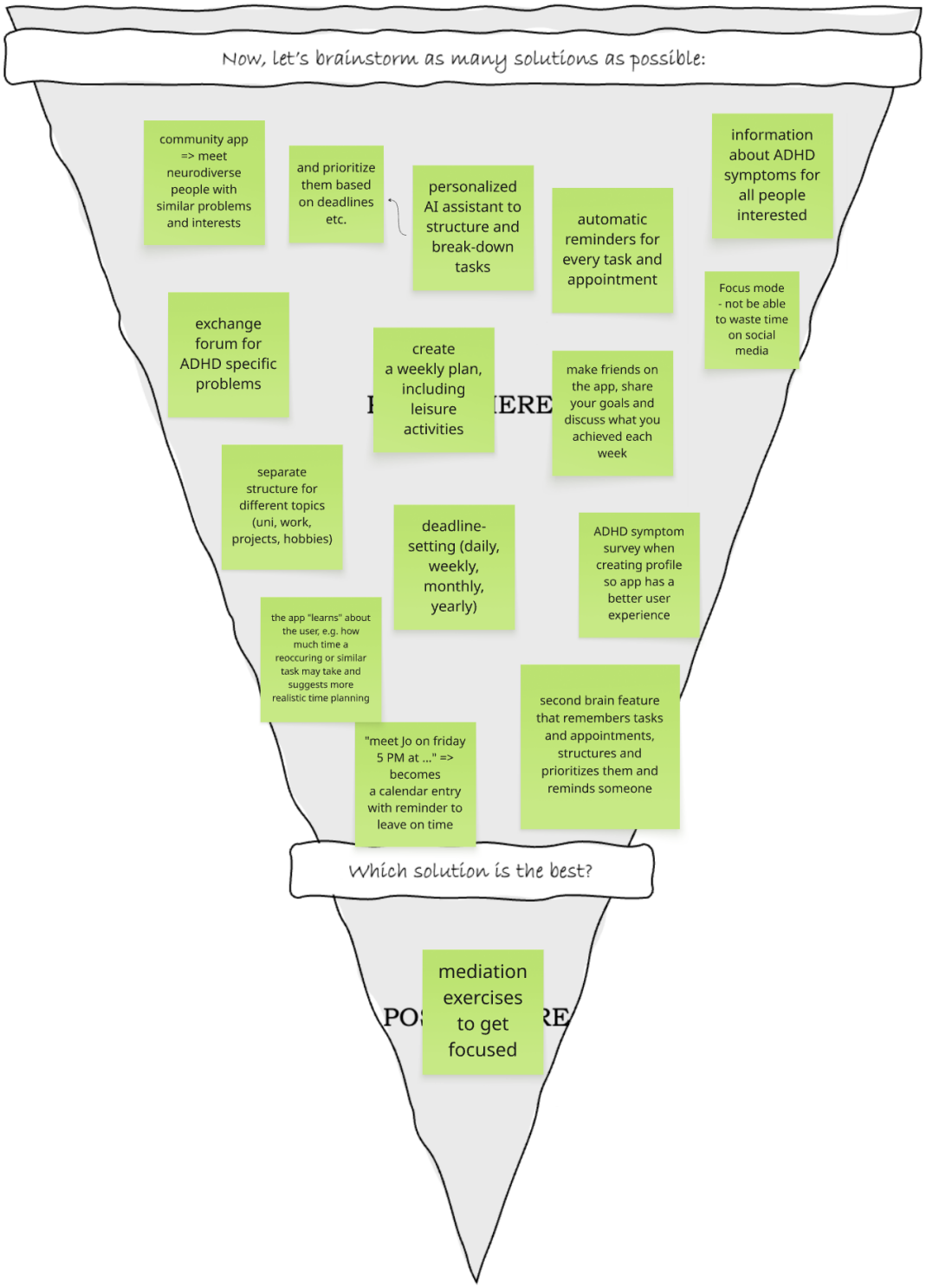


Figure 5: Activity of the Ideation Process

3.4 Critical Reflection on Design Thinking

What worked well with this approach for your project?
Empathise went really well.
We had a big advantage in participant recruitment through our teammate who is dealing with ADHD herself. She knows and regularly

meets neurodiverse people, so she used her network to talk to as many as possible about our project. This also brought a strong sense of authenticity and relevance to our user research, as our interview style therefore matched with all those affected. We were also lucky to have interviewed people with very diverse backgrounds, almost none of them are from computer science. This helped us avoid the usual tech-focused bias and let us better understand actual, everyday needs and emotional challenges of those affected. As a mostly neurotypical team, we genuinely enjoyed listening to the interviews! It helped us get an understanding of what it actually means to have ADHD (also because we talked a lot to our teammate about this topic apart from the interviews). We wouldn't have gotten those insights by just reading online articles.
Define became much easier after the first few interviews.
Having heard about real issues and stories helped us identify recurring needs and struggles, which we clustered and used for our user personas. It also gave us a deeper purpose for our project, because we dealt with real people.
Ideate was one of the richest parts.
After getting a grip on the real issues, brainstorming almost happened automatically. We came up with so many ideas and discussed very different directions - not only in the beginning of the project, but over the entire time period. There were so many possibilities and paths we could have taken - but we really went into it and repeatedly discussed what really mattered most, so we aligned all our ideas to three core functions for our app that felt both useful and doable.
Prototype was the fun part!
Because we had so much input from our interviews and a strong sense of direction from our ideation sessions, it was easy to translate our ideas into first drafts. We quickly moved from paper sketches to digital mockups in Figma. The process felt intuitive, and we really enjoyed bringing the ideas to life - especially when the first screens became interactive and our app became more real.

What challenges did you encounter? (time constraints, participant recruitment, conflicting feedback)
One major challenge was the feeling of being stuck in an endless loop.
Finding a definition of done in an iterative process is difficult, especially in combination with a complex topic like ADHD - there is no "perfect" solution that fits all, there is no "perfect" functionality, there is no "perfect" design. We often found ourselves questioning whether we were "done" or if we should go back and revisit earlier phases. During prototyping, testing, and team discussions we kept generating new insights that challenged our earlier assumptions, which made it hard to reach a "final" version. But we started to realise that it doesn't have to be perfect - it just has to be done with thought. Perfect wouldn't be as lovable as the effort we put in.
We also never really left the ideation phase.
There were just so many interesting features we could include, and we got a bit overwhelmed by that. Maybe also classic ADHD overthinking on one part. Our app idea became very complex very quickly, and it was difficult to narrow it down to the essentials. This also made the prototyping phase more intense than expected.
Testing was a bit tricky.
While recruitment was easy, most of our interviewees were not experienced with productivity tools or app design in general. Because of that, their feedback was sometimes superficial or limited to first impressions rather than usability and feature depth. Also some testers had a hard time focusing during the sessions or did not even respond, which may have been due to ADHD itself, making it hard to collect consistent and usable feedback.

When did design thinking not provide clear direction?
Design thinking did not guide us clearly when it came to prioritisation.
Because of the explorative and iterative nature of the design thinking process, we almost generated too many ideas - it was difficult to decide on which potential features were essential and which ones we could leave out. The design thinking process did not really offer a clear tool for narrowing down ideas in a user-centred yet feasible way.
Design thinking did not guide us clearly when it came to testing.
We weren't sure how to interpret vague or limited feedback, especially when users were not as engaged as expected. Also, sometimes users had different opinions or ideas about certain details. The design thinking process is about improving based on feedback, but it doesn't give clear instructions when feedback isn't actionable or contradictory.

What alternative approaches might have been more effective?
We did some research and found a structured prioritisation method, called the MoSCoW method (Must-have, Should-have, Could-have, Won't-have), which probably would've helped us during ideation and definition phases to stay focused and avoid overloading or constantly changing the prototype.
In terms of testing, using standardised usability tasks could have provided more actionable and specific feedback, especially since many of our testers were unfamiliar with productivity apps and the concept of early-stage prototypes.
However, those methods could have complemented the design thinking process, but we don't think they would have been a good substitute. We all agree that the process itself is a great user-centred method, especially because it constantly brings the focus back to real users and their needs. None of the other methods we have used so far or learned about before put this level of emphasis on empathy - having run through the design thinking process for the first time, we now truly understand its value and how much it can shape the direction and quality of a project.

4 Requirements

In the following we want to show our learning journey through changing requirements during our project.

4.1 Stage 1

Functional requirements Our initial app concept features five core functionalities: Mediation, Connect, Task (To-Do lists), Map of your mind, and Declutter.

- **Task**: with deadline and category
- **Map of your Mind / Declutter**: Talking about topics that are on your mind
- **Second Brain**: a quick way to write down tasks and thoughts so they are not forgotten
- **Connect**: to interact with others who have ADHD
- **Mediation**: as a balance, to be able to focus better afterwards

Non-functional requirements

- **clean look**
- **minimal text**
- **no gamification**
- the appearance can be **customized** to meet individual needs

The functional and non-functional requirements were developed during a very early brainstorming phase, as illustrated in Figure 6.

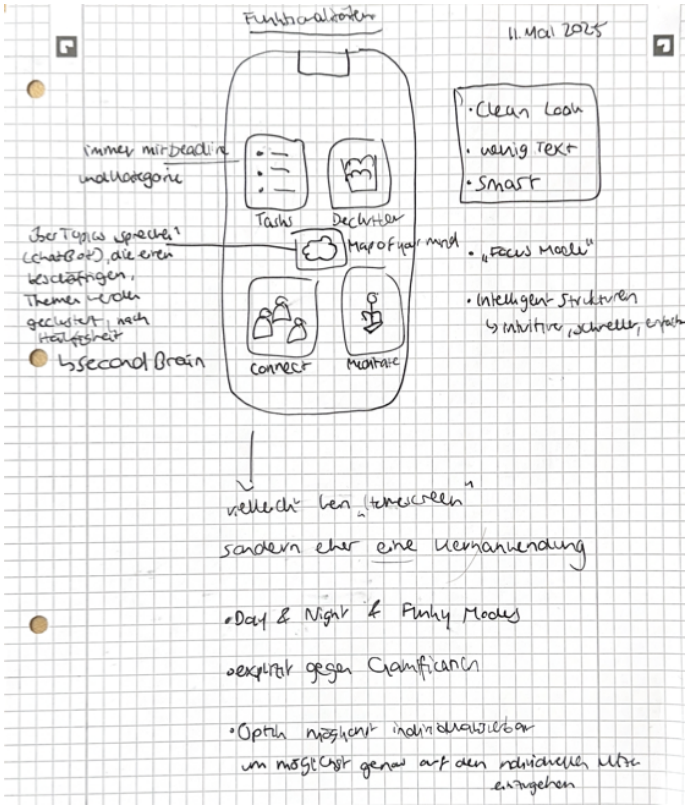


Figure 6: Initial Concept Brainstorming

4.2 Stage 2

- **Connect**: A community app to make friends, share goals and achievements
- **Intelligent lists**: large tasks are automatically broken down into smaller subtasks, tasks are prioritised based on importance, users are reminded of tasks -> second brain
- **Information hub**: Information, tips, and tricks about ADHD that can help assist in coping with it
- **Focus Mode**: hides / disables other apps to avoid distractions, such as those from social media

Added
The focus has shifted somewhat from a productivity app to a community app. The app is now intended to have a core that revolves around community, where the focus lies on social interaction. This especially includes finding friends and sharing goals and achievements, like completing a difficult task, with them. This is meant to foster mutual motivation when working on one's own task. Overall, this requirement has emerged from the insight that people with ADHD find it more comfortable and easier to share their struggles with others who are also affected, as they speak from shared experience and truly understand the challenges involved. In addition, an ADHD knowledge hub will be included. It is like a catalogue with tips and coping strategies designed to help those affected manage ADHD more effectively. Another new feature is the Focus Mode, which hides other apps, as we identified the phone itself as a major source of distraction, and we it to be part of the solution.
Modified
The task list has been modified: because we identified users having trouble doing so, the lists should become smarter, with tasks being broken down into smaller steps and assigned deadlines to remind the user. This should save the user time and effort in managing everything. The concepts of decluttering / map of mind are now part of a central 'Second Brain' feature.
Removed
Meditation as a core feature has been removed. However, since it is still considered something useful, meditation tips and guides are included in the ADHD knowledge hub.

4.3 Stage 3

For the requirements of Stage 3 we want to refer to the following chapter 5. In our opinion there is a description of all requirements that have been current in the past or still are current until now and how and why we have changed the requirements over time.

5 Design

In the following, we want to outline the evolution of our design process, from initial paper sketches to the final digital prototype. We highlight key points of iteration, explaining what feedback or insights led to major changes. Usability testing results are summarized, focusing on user challenges and our responses. We also explain important design decisions regarding layout, navigation, and visual style, and reflect on concepts we explored but ultimately discarded, including the reasons behind those choices.

5.1 Mockups/Low-Fi Prototype

Figure 7 to 11 provide an overview of the initial app concept and design, illustrated in the form of sketches. The Figures 7 to 9 illustrate the login and account creation process, designed to be minimal and intuitive, and needs to be completed only once. Following the onboarding, Figure 10 shows the Focus Screen, offering access to the AI Assistant Neura and an task overview. Ultimately, Figure 11 outlines the overall navigation structure and the three core screens: Focus, Connect and Browse

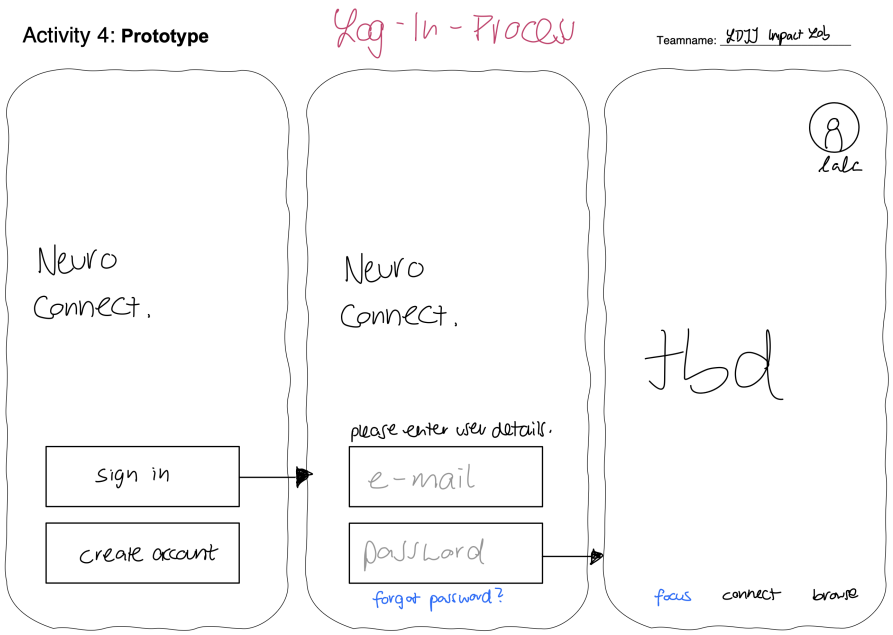


Figure 7: Initial UI Mockup Log-In Interface

Figure 7 presents the login screen, offering options to either sign in or create a new account. A central design principle of this screen is to make the login process as uncomplicated as possible to avoid initial barriers to entry. Accordingly, users are only required to log in occasionally, for example after an major app update.

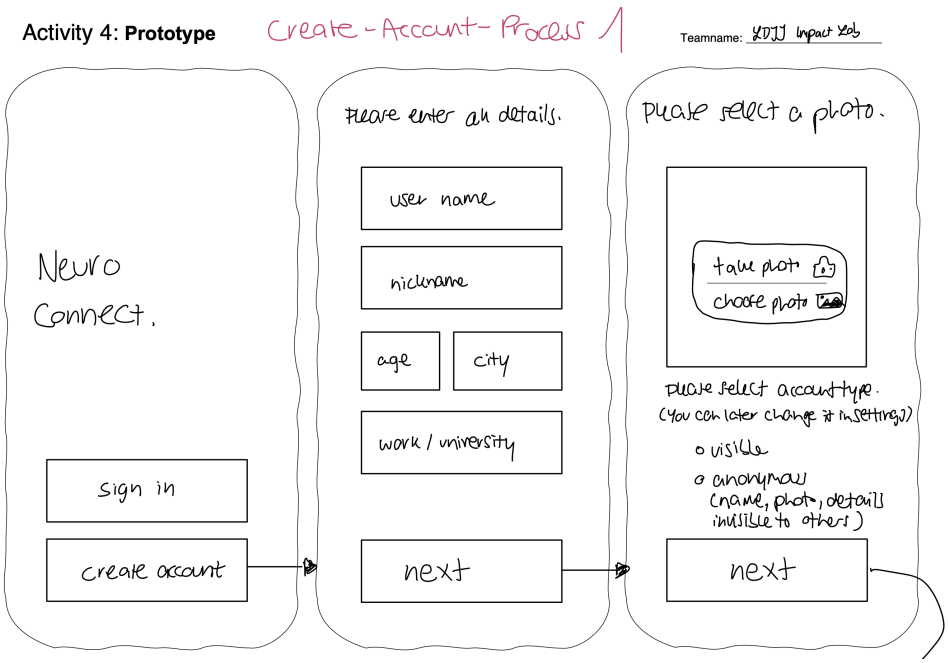


Figure 8: Initial UI Mockup Profile Settings

Figure 8 shows the first screen of the account creation process, which is only required when creating a new user account. Basic personal information such as age, city and profession along with the selection of a profile photo are requested here. The user is also given the option to define the visibility of the account (name and photo) in social posts and choose between a visible or anonymous profile. The data provided in this phase is crucial for the app's social function and enables personalised interaction within the community.

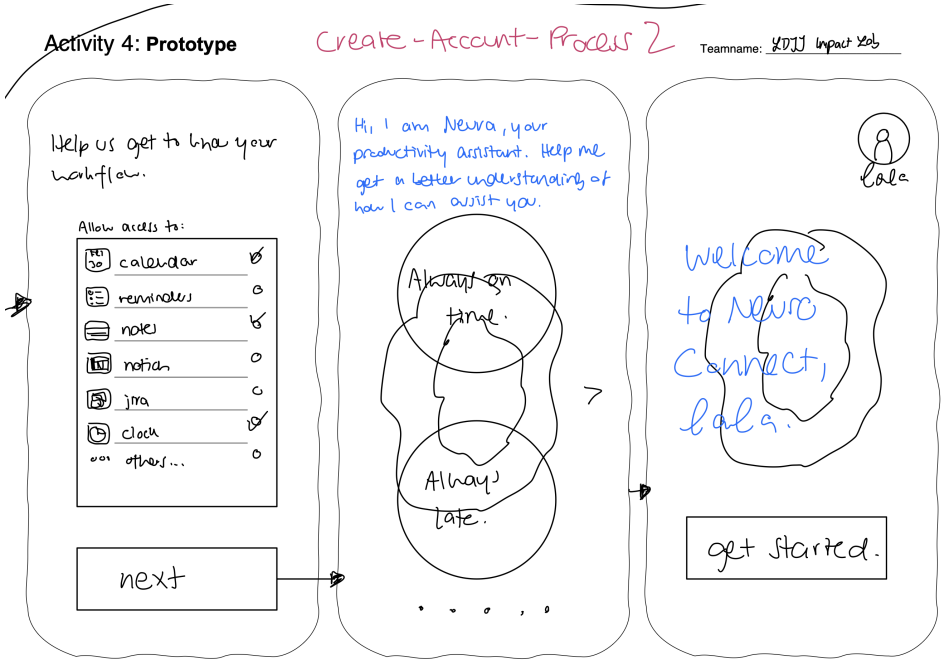


Figure 9: Initial UI Mockup System Permissions and Initialising Personalisation

The second step of the onboarding process is shown in Figure 9. On this screen, read and write access to other applications such as

the user's calendar, reminders or notes is requested. These accesses are essential to provide the best possible app experience, but each user is in control over the extent of data access. In addition, the AI integrated AI assistant, Neura, asks for information in order to better adapt to the user's individual needs and habits and thus offer personalised support.

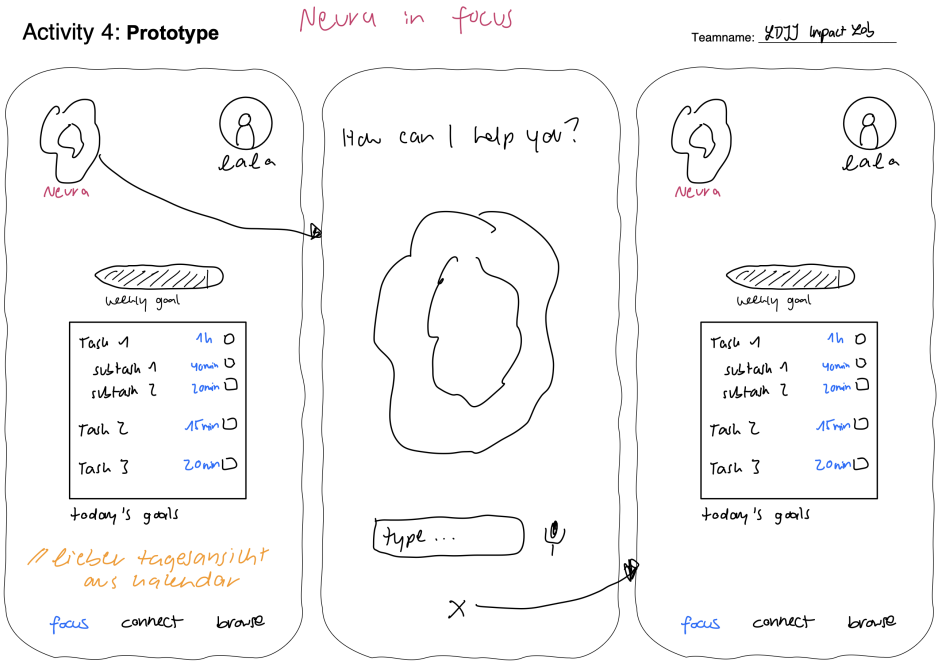


Figure 10: Initial UI Mockup Neura

One standard app feature, the so-called *Focus Screen*, designed to support the user in terms of productivity, is presented in Figure 10. The focus screen also provides an overview of the upcoming tasks. Additionally, in the upper left corner, a quick access to "Neura" is provided. This quick-access feature to Neura is designed with neurodiverse users in mind, particularly those affected by ADHD who often have rapidly changing or intrusive thoughts that can interrupt working on current tasks. By offering a straightforward way to externalise such thoughts immediately, the interface enables to quickly to avoid mental distractions that would break their focus.

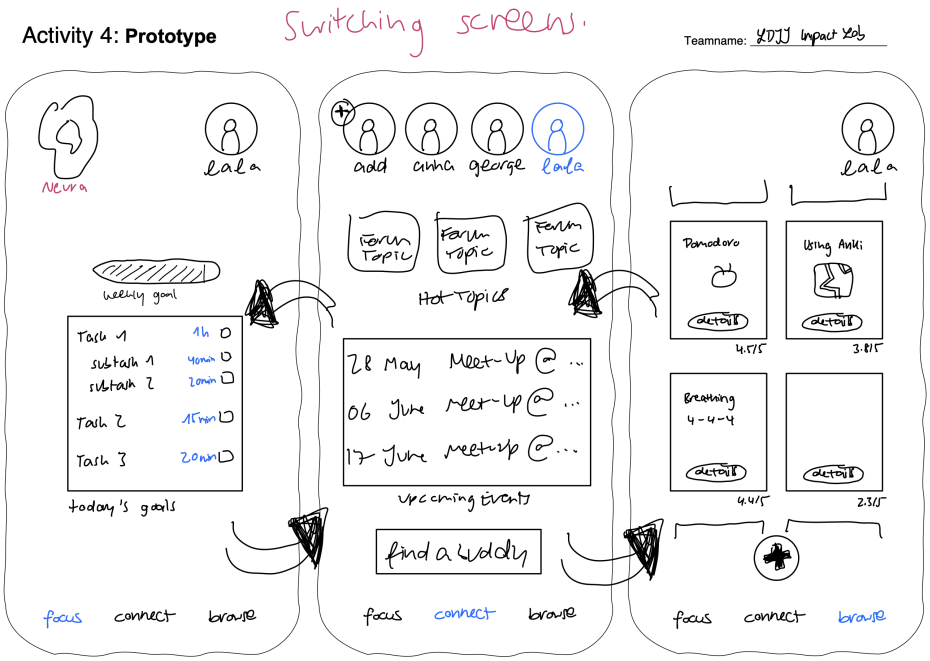


Figure 11: Initial UI Mockup Overview Screens

Figure 11 outlines the additional main screens of the application: *Connect* and *Browse*. These screens can be accessed either by a swipe gesture, to reach the adjacent screens, or at any time via the corresponding buttons in the lower navigation bar. The *Connect* screen facilitates interaction with other users and social features. It displays upcoming meet-ups in the vicinity of the user and provides access to community forums on various topics. Furthermore, the *Browse* screen allows user to explore educational content, including articles that present coping strategies for managing ADHD.

5.2 MLP/High-Fi Prototype

Figures 12–15 show the first high-fidelity (high-fi) prototype of **NeuroConnect**, translating the paper sketches into flawless visual screens while introducing several interaction layers that were only conceptual in the low-fi sketches. What was not clear from those initial sketches was the design language: here we settled on a light, muted blue-turquoise on an off-white with a slight beige tint background. This combination is not only visually appealing but also offers significant practical advantages. Blue is often associated with calmness, trust and professionalism, aligning well with NeuroConnect's purpose. Furthermore, the colour combination offers clear contrast and maximize readability. The light, not saturated colours, reduce visual noise and do not cause overstimulation.

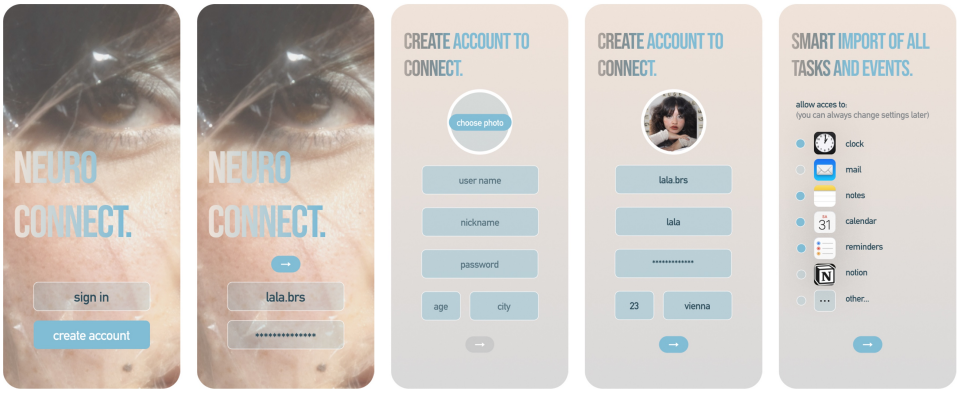


Figure 12: First iteration High-Fi Prototype Login and Setup Screens

Figure 12 illustrates the login process of the first high-fi prototype. You can now see the app-wide design that we have decided on. The biggest change here is that the option to choose between a visible and an anonymous account has been removed. The other

changes have been rather minor. Firstly, a background image has been added to the login screen to round off the design. secondly, the design of Neura has been properly visualised for the first time, i.e. what it looks like when you use Neura.

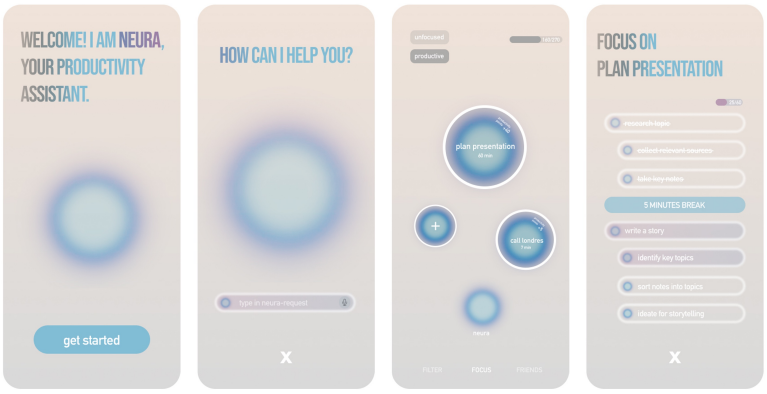


Figure 13: First Iteration High-Fi Prototype Neura and Focus Screens

As shown in Figure 13, the focus page has undergone the biggest changes. The tasks are no longer arranged in a kind of list where you can tick off completed tasks. Tasks are now displayed in bubbles. The bubbles display the expected duration of the task and also productivity points, which have also been newly introduced. The new representation in bubbles makes it possible to visualise the size and importance of the tasks by the size of the individual bubbles. The productivity points are another new feature on the screen, as users can create weekly task goals, not the number of tasks but in terms of productivity, which is measured by productivity points. There are now also two buttons, unfocussed and productive. If you press productive, you are saying that you are ready to start a task. However, if you are unfocussed, the user is given support to focus, for example through guided meditation exercises, so that they can then continue working on their tasks. Furthermore, a significant addition is the new lock screen, which now appear when the user begins a task. This screen effectively provides a schedule of the tasks, offering a clear overview of the current progress within the workflow.

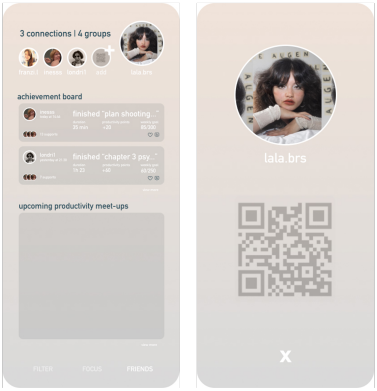


Figure 14: First Iteration High-Fi Prototype Connect Screens

Figure 14 presents the redesigned connect screen. The main change to the connect screen is that the forums have now disappeared. The reason for this is that they were felt to be too distracting because it is easy to get lost in them and get stuck reading somewhere. Instead of the forum topics there are now achievement boards. These are related to the new feature of the productivity points. The achievement boards show what friends have done during the week or as the last task. This creates a little pressure, which is intended to stimulate your own productivity and motivation to complete tasks.

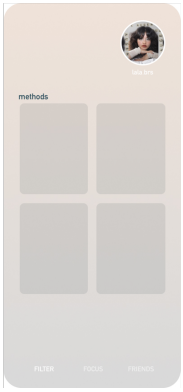


Figure 15: First Iteration High-Fi Prototype Browse Screens

The browse screen, shown in Figure 15, remains consistent in design and function, providing an overview of different coping strategies for ADHD.

Table 1: Summary of Main User Feedback and Corresponding Design Changes from Low-Fi to First High-Fi

User Feedback	Implemented Design Change
The initial task list felt too rigid and was not visually helpful in prioritising tasks.	Introduced bubble-style visualisation for tasks with size indicating their size and importance, allowing better visual prioritisation.
Forums were perceived as too distracting and not in line with the purpose of the app.	Forums were removed.
The selection of an anonymous account does not fit with the idea of the connect feature.[1]	The anonymity toggle was removed to avoid giving users the chance to create an anonymous profile.
It has been reported that a little pressure often drives people to start tasks rather than procrastinate	Introduction of the productivity points system with productivity goal-setting and tracking the weekly productivity progress of friends and which tasks they have recently completed.
Users reported that they sometimes feel lost within a task and lose track of things while completing the task	Added a lock-screen that display the current task progress, showing completed steps, the current step and upcoming steps to maintain orientation.

5.3 Intermediate High-Fi Prototype

In the following we present our intermediate prototype, this step is the result of our second feedback round and our planned visual and functional improvements to the app.

[1]This point was not based on user feedback but was a deliberate team decision made during the high-fi implementation. This is included to explicitly detail the design decisions and ensure their clarity and transparency.

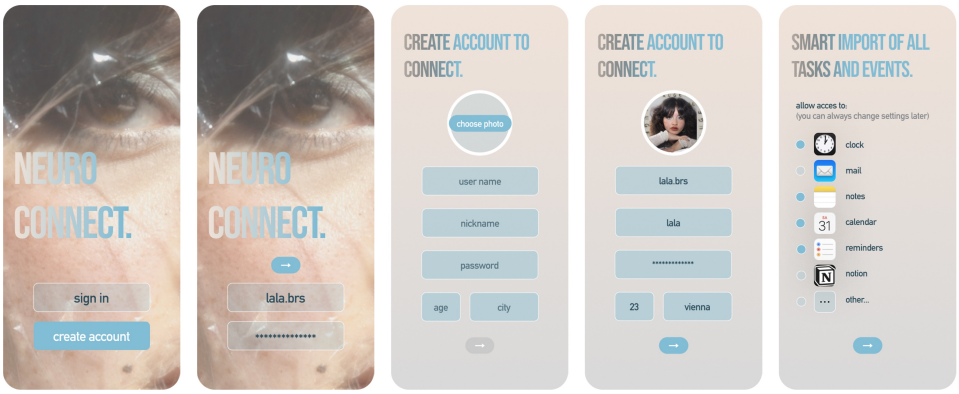


Figure 16: Intermediate High-Fi Prototype Login and Setup Screens

Figure 16 presents the unchanged login & set-up flow. After the second round of feedback we decided to leave these screens exactly as they were, because all five testers moved through the steps without hesitation; no additional guidance appeared necessary at this stage.



Figure 17: Intermediate Iteration High-Fi Prototype Neura and Brain-Dump Screens

In Figure 17 we illustrate how we fleshed out the *Neura* assistant. The feature now serves as an anchor point that ingests every piece of information the user supplies, organises it into a dynamic brain-map, and allows the map to be filtered by upcoming deadlines via the control bar at the bottom of the screen. This tight integration bridges Neura, *Focus* and the brain-dump flow.

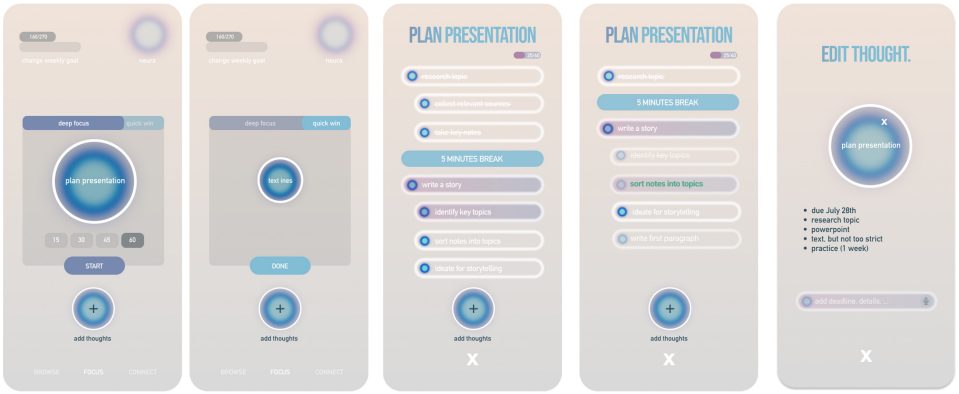


Figure 18: Intermediate Iteration High-Fi Prototype Focus Screens

Figure 18 shows the expanded *Focus* module. We enlarged the feature set enough to justify a dedicated tab next to Neura while still keeping both areas closely connected. The screen now offers two pre-selected task modes: Deep Focus proposes a single, high-impact

task for an extended concentration session, whereas Quick Win surfaces a lighter task for moments when intense focus feels unrealistic.

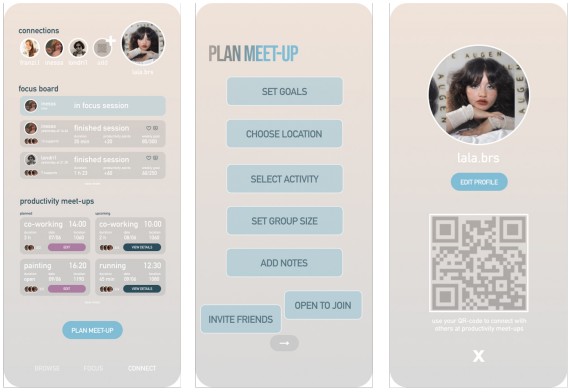


Figure 19: Intermediate Iteration High-Fi Prototype Connect Screens

Figure 19, shown next, details the updated *Connect* screen. Users can browse planned meet-ups and schedule their own events. Guided by earlier feedback about distraction, we kept the colour palette muted and the information purely informative. Security concerns, especially around anonymous users, remain on our radar; corresponding safeguards will be designed in the upcoming iteration.



Figure 20: Intermediate Iteration High-Fi Prototype Browse Screens

Figure 20 presents the revised *Browse* section. Beyond listing coping methods for ADHD, the screen now includes peer experience reports. These first-hand stories help users discover strategies in a more organic, friend-to-friend way. Initially, we had removed the forums based on concerns about distraction. However, later user feedback expressed a clear wish for more dynamic and authentic exchange of personal experiences. This feedback was considered convincing and led to a reassessment of the earlier decision. As a result, the forum-like feature is reintroduced to offer users these kind of experiences.

Table 2: Summary of the Main User Feedback and Corresponding Design Changes from first High-Fi to second High-Fi

User Feedback	Implemented Design Change
The focus screen, brain dump, and Neura were not very detailed in the first iteration, so their exact use and functionalities did not become immediately clear.	We further differentiated and detailed the brain dump feature, to now not only feature the input capability but also a more detailed an interactive representation of the saved tasks.2095102
The functionality of a focus session was missing from the prototype, so users were wondering what to do with a task once it was saved into the app.	To mitigate this, we introduced a dedicated focus screen, differentiating quick wins and deep focus sessions, and provided detailed task structures.
For the connect functionality, testers were asking for a more intuitive design on the focus board.	We redesigned the focus board to be more intuitive and less confusing.
The productivity meet-ups feature was missing in the previous draft.	We drafted an intuitive overview for the meet-ups feature, keeping in mind the previous feedback.
Users were asking for a more dynamic way to share experiences, not just reading static strategies, but actually sharing and reading about real experiences and best practices.	Following this feedback, we designed the browse feature around two main sections: the recommended methods catalogue and ADHD experiences shared by others as best practices and tips.

Looking ahead, For the final usability round we will once again invite five adults living with ADHD and ask them to “play new user” while we observe. The script will stay deliberately lightweight so we can see how far participants get before asking for help. After a brief welcome chat we will hand over the phone and instruct them to (1) sign up or log in, (2) link at least one calendar in the data-import step, (3) open *Focus*, perform a brain-dump, then edit one of the resulting thought-bubbles, (4) switch to *Browse* and find a productivity technique, and (5) jump into *Connect* to send a friend request and start organising a meet-up. At this stage every feature is visually present, although not all elements are fully clickable yet.

5.4 Final Product

In the following section we describe the final stage of our prototype based on the last iteration of user feedback. This includes a description of the functionality shown on the figures as well as a few features that go beyond.

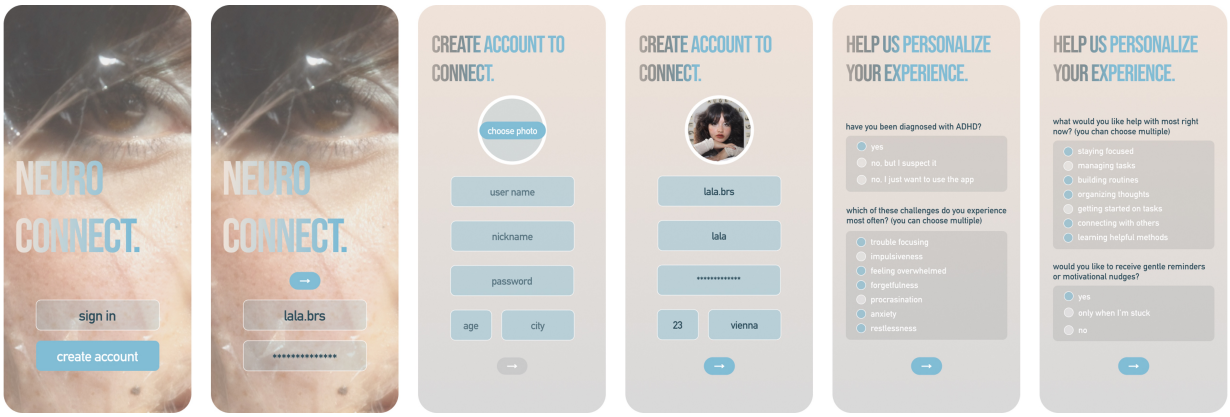


Figure 21: Final MLP Prototype Login and Setup Screens

When opening the app, new users have the option of creating an account and existing users can log in with their user name and password (screen 2). Screens 3 to 6 show the registration process. The user must assign a user name, a nickname and a password. They can also enter their age and place of residence and complete their profile with a profile picture. The new user is then asked a few questions so that an initial rough personal profile of the user can be created, which will be used to improve and personalise the user experience over time. The first question asks whether the user is affected by ADHD, assumes that they are affected or simply wants to use the app as it is. Next, the user is asked about their most frequent challenges. These include the problem of being able to focus, forgetfulness and procrastination. The last personal question refers to the things in which the user would most like or need support. The user can choose from the following, for example: staying focused, managing tasks, building routines, organizing thoughts, getting started, connecting with others and learning helpful methods. Finally, the user is asked whether they would like to

receive notifications and nudges. They can either authorise or prohibit this in full or restrict it to sending messages and notices only if Neura determines that the user requires assistance.



Figure 22: Final MLP Prototype Neura and Brain-Dump Screens

After registering and setting up the app for the first time and after logging in, the Neura start screen awaits the user. With a swipe across the screen from bottom to top, the user accesses the input field, which allows them to record their initial thoughts. Thoughts can be entered both by text and by voice input. If the user has nothing on their mind that they want to write down in the app, they can go directly to the focus screen via the 'X' at the bottom of the screen. (More information on this follows in the next section). If, on the other hand, the user has entered all their thoughts or simply wants to see an overview of the thoughts they have already collected, they can click on the 'view all thoughts' button. This is followed by an overview of all the user's existing thoughts, tasks and ideas. The user can filter the views of thoughts according to temporal relevance. Screen 4 shows an example of the tasks and thoughts that are relevant for the current week. The user can click on the individual thoughts to edit and categorise them quickly and easily in a detailed view. Thoughts can either remain in the general thought overview or be converted into 'Tasks' for the Focus Screen, 'Plans' for the Connect Screen or 'Experiences' for the Browse Screen (screen 5). The user can also display a detailed description of the thought. For the 'plan presentation' task shown here, Neura generates a detailed plan to complete the task step by step. The thought input bar remains available on all screens to ensure that thoughts can be captured at any time so that the user can concentrate on the current activity. In the same way, they can return to the overview of all thoughts at any time if they want to complete or simply edit another train of thought. If the user categorises the 'plan presentation' task as a 'focus task' and clicks on the corresponding button, screen 2 of Figure 23 would be the one displayed. However, this screen would then contain pre-filled fields. For example, the first field would already contain the title of the task 'plan presentation' and the second field would contain the name of the subject in which the presentation has to be given.



Figure 23: Final MLP Prototype Focus Screens

Figure 23 shows the Focus Mode screens. The first image shows the centre screen of the mode, which can be seen from the white 'Focus' at the bottom of the screen. These buttons allow the user to switch between the three screens of the main functions 'Browse', 'Focus' and 'Connect'. Alternatively, they can swipe to the left or right to go to 'Browse' or 'Connect'. The blue bubbles represent the different tasks, the size the effort required for them. The bar in the top left-hand corner shows the progress a user has made so far.

The user can also decide, for example, whether to display the tasks for the current day, the current week or the whole month. If the user has thoughts in mind that they want to record, they can use the button in the top right-hand corner to go to Neura, where they can write them down. Using the blue ‘add session’ button, the user can create a new ‘Focus session’ and add it to the overview.

Screen 2 shows this procedure. The user can give the new session a name and assign a purpose, specify the time period over which the session is to be performed and how many minutes per day or week they would like to work on the session. The user can also define initial interim goals, specify how long they want to work on the first interim goal and whether the app should incorporate smart breaks. Using the ‘add friends’ button in the top right-hand corner, the user can add friends to this session if they are involved in group work, for example, or if the user simply wants to invite friends to co-working.

The third image shows the view if the user in the example here had clicked on the bubble with the task ‘Computing for social good’. The progress of the selected task is shown in the top left-hand corner. In the lower half of the screen, the user sees the purpose of the task, their previous successes, which they can share with others if they wish, and their latest notes on this task. In the upper half, the user can define the next intermediate goal and the time they want to work on and for how long, and start a focus session by clicking on the ‘Focus’ button.

Figure 4 shows the screen view during such a focus session. A timer runs in the top left-hand corner, showing the remaining time of the session. Below this, the user is shown the interim goals that they have previously formulated and can now mark as completed with one click. In addition, they can add further things at any time that they have got done additionally and had not yet formulated. In the centre of the screen, the user is permanently shown the purpose, goal or reason for the task as motivation. In the window in the lower half of the screen, the user can enter thoughts at any time and thus record them for the future. If the user wants to end the session before the predetermined time has elapsed, they can click on the timer. Screen 5 shows the scenario described below. Before the session ends, the user is asked again whether they really want to end the session and is shown the time they have spent in the focus session up to this point. If the user decides to continue the session after all, they will return to screen 4. When they end the session, they are taken to screen 6.

Here the user is shown what they have achieved in the past session and at the same time what is still outstanding. They can share their successes with their friends again at will using the ‘share’ slider. The user can also see the thoughts they wrote down during the session. If required, the user can immediately formulate and start the next session. If they do not do this and leave the screen with the ‘X’ at the bottom of the screen, the app takes them back to the overview of their tasks, which is shown in Figure 1.

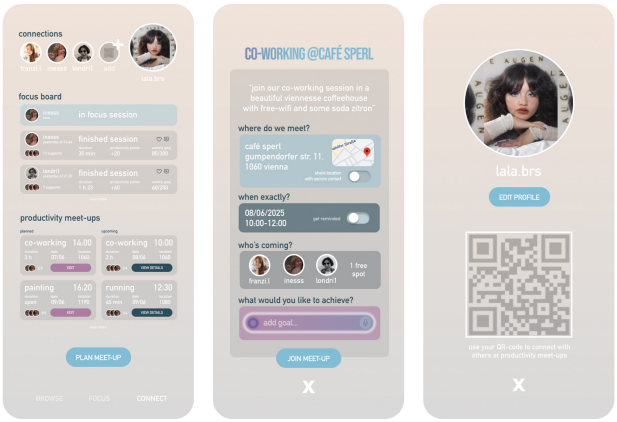


Figure 24: Final MLP Prototype Connect Screens

Figure 24 shows the screen of the second main function ‘Connect’ in the first image. At the top of the screen, users can browse through their friends and access their profiles, similar to how users of the ‘Instagram’ app can browse through stories. The user’s own profile picture is displayed in the top right-hand corner. Clicking on this takes the user to the third screen of Figure 24. They can use the ‘edit profile’ button to edit their own profile or show friends the QR code, which they can scan to arrange to meet up via the Neura app.

Below the friends’ profiles, the user is shown an overview of their activities. Completed focus sessions are listed here if the users have activated the ‘Share’ slider described above. Users are also shown when one of their friends is currently in a live focus session. In future, it is planned to implement the option for users to join such sessions or to ask the person concerned whether they can attend this session in the form of body doubling. Friends’ past focus sessions can be commented on and/or ‘liked’, and you can also see how long a friend has been in a focus session and how many productivity points they have collected with it. Mutual encouragement as well as an insight into the productivity of others is intended to increase one’s own motivation.

The bottom half of the screen shows an overview of planned productivity meetings. The meetings are each labelled with a name, a time, a date, a duration and a location. In addition, there is a maximum number of people who are planned for the meeting and the number of people who have already confirmed their attendance. You have already confirmed your attendance at the meetings listed on the left under ‘planned’ and can edit your confirmation using the ‘edit’ button. The meetings under the ‘upcoming’ category on

the right-hand side are, as the name suggests, taking place in the future and you can still join them as the maximum number of participants has not yet been reached. You can join by clicking on the ‘view details’ button.

This takes the user to the second screen in Figure 24. From top to bottom, this first shows the title of the planned meeting. Below this is a short description of the meeting of the initiator, followed by the address. The address can be shared with a security contact stored in the profile using the slider on the right, so that someone knows where you are if you are meeting new strangers or at an unknown location for the first time. The date and time can also be seen on the screen, and the user can use the slider next to it to have the app remind them of the appointment. Finally, they can see the profile pictures of the people who have already confirmed their attendance at the meeting and have the opportunity to state what they would like to get out of the meeting or what they would like to do at the meeting in relation to the example of a co-working meeting shown here.

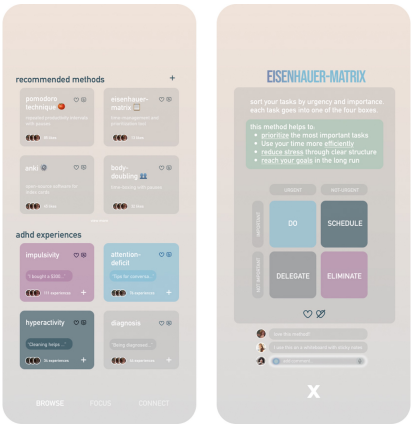


Figure 25: Final MLP Prototype Browse Screens

Figure 25 shows the last of the three main functions of our app: ‘Browse’. Here, users can browse through a kind of catalogue containing various methods that help them to be productive, prioritise tasks or find a balance in their everyday lives. The Pomodoro technique, the Eisenhower matrix and body doubling are shown here as examples. But they can also find suggestions for meditation or sport here. The user can see how many other users have given a method a like. If the user selects a method and clicks on it, he will receive more detailed information on this method. This can include, for example, the structure and functionality of the method. This is shown on the right-hand screen. Here the user also sees comments on the method from other users and can share their own experiences so that other users can draw on them. With the ‘+’ in the top right-hand corner, users can add their own methods if they are not yet included in the catalogue.

In the lower half of the ‘Browse’ screen, the user finds experiences from other users on the three main characteristics of ADHD: hyperactivity, impulsivity and attention-deficit. There is also the topic of ADHD diagnostics. All users can share their experiences here and discuss them in the form of approval with likes or comments.

Table 3: Summary of Main User Feedback and Corresponding Design Changes from Second High-Fi to final MLP

User Feedback	**Implemented Design Change**
Users wanted clearer guidance on what to do after recording a thought in the brain-dump, the process felt incomplete.	Added the ability to categorize thoughts directly after recording them and to directly convert them into Tasks, Plans, or Experiences
The task view in Focus Mode lacked context and didn’t allow enough interaction or structure around task execution.	We added structured Focus Sessions with progress tracking, timers, goals, motivational prompts, and break planning.
There was no clear structure for creating, viewing, or joining productivity meet-ups.	So we designed a complete meet-up scheduling system, including name, description, location, participant caps, confirmation tracking, and reminders.
The Browse section was too static and felt like a simple list of methods. Users wanted to engage with real experiences.	Expanded Browse to include detailed user-submitted experiences and comments, plus the ability to like, comment, and submit new productivity methods.

6 Architecture

In the following, we want to explain why we chose a web app instead of a mobile or desktop app, based on what we learned from users. We also show how the different parts of the app work together in a system diagram. Then, we describe what kind of data the app uses and how this data is organized. We outline the typical steps users take while using the app, and we list the external services the app needs, like APIs or login systems. Finally, we briefly explain how the app can grow and handle more users in the future.

6.1 Platform justification

We decided to design a mobile app due to the fact that the smartphone was often mentioned as the main distraction factor in the interviews. What's more, nowadays people have their smartphones with them almost all the time, so there is always the option of using the app if, for example, they want to enter a new appointment, a new reminder or a new task to be completed or simply check what else is on the agenda for the current day. The ability to get in touch with new people in the style of a social network also speaks in favour of a mobile app. A smartphone is essential for the functionality of making appointments with other people and meeting them on site for any navigation that may be required.

6.2 System components diagram & Data architecture

In the following, we provide a quick overview of the internal architecture required to to make this app a reality, and help potentially tens of thousands of user in their daily lives. At the centre lies the **center**, which hosts the three feature modules, Focus, Browse and Connect, plus the Neura assistant that ties them together. Every call reaches the core through a REST API protected by an authentication layer that issues short-lived tokens, ensuring each request is properly identified and authorised. A notification service listens to domain events and delivers push or e-mail reminders at the right moment. Beneath these services, a relational database keeps the single source of truth for structured data, while a vector index stores high-dimensional embeddings and an object store holds larger binary assets, each in the same GDPR-compliant cloud region.

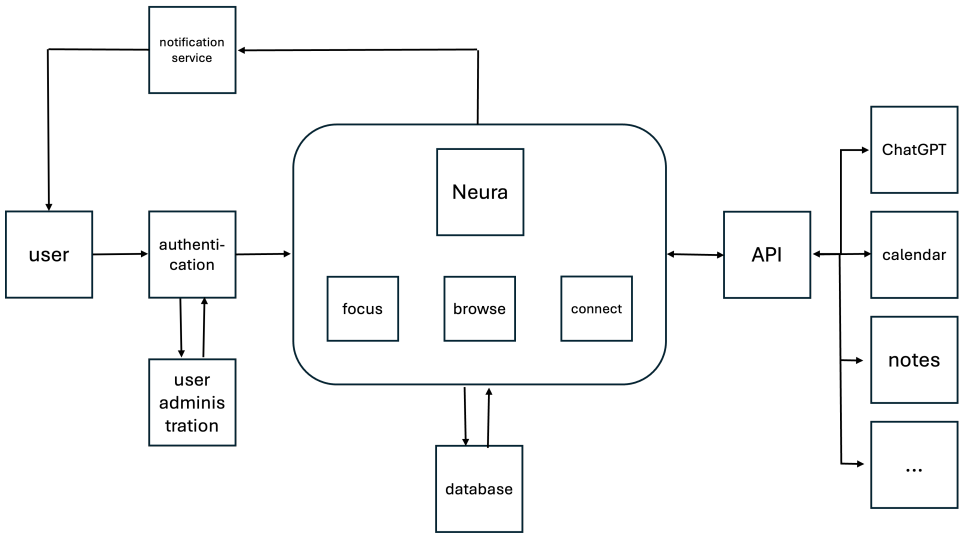


Figure 26: Data-Architecture Draft

All external traffic passes a single backend API, which rate-limits calls, strips personal identifiers and relays only the minimum data needed to external calendars, note apps and the ChatGPT endpoint. A thought entered in Focus is first turned into an anonymised embedding, stored in a local vector index, and only its most relevant snippets are sent to ChatGPT; the response returns through the same gate and appears in the UI. A server-side notification loop pushes reminders via a queue-driven worker, so alerts respect quiet hours without keeping the app open. Data lives on EU-serves: a relational database for accounts and meet-ups, a vector DB for embeddings, and object storage for images, keeping search fast and the storage inexpensive. Also Raw text should never leave this environment. only hashed or embedded fragments travel outward. Last but not least, a self-service portal (still to be implemented) lets users export or erase their data at any time, underscoring our core pledge: **your information stays private, and Neura sees only what it needs to help.**

6.3 User flow documentation

If a user is using the app for the first time, they must first create an account with a user name and password. This is followed by a few initial questions from the artificial intelligence, such as ‘what do you expect from this app?’, ‘what stops you from being productive?’ or ‘what do you like to do in your free time/what motivates you?’, which it uses to create a profile of the user and offer personalised help. If the user is already registered and opens the app again, the artificial intelligence Neura asks them questions such as ‘How are you today?’ or ‘What’s on your agenda today?’. These questions can also be skipped so that the three main functions Connect, Focus and Browse can be used directly. You can access Neura from anywhere in the app and use it to communicate thoughts, tasks or appointments, which it then processes accordingly.

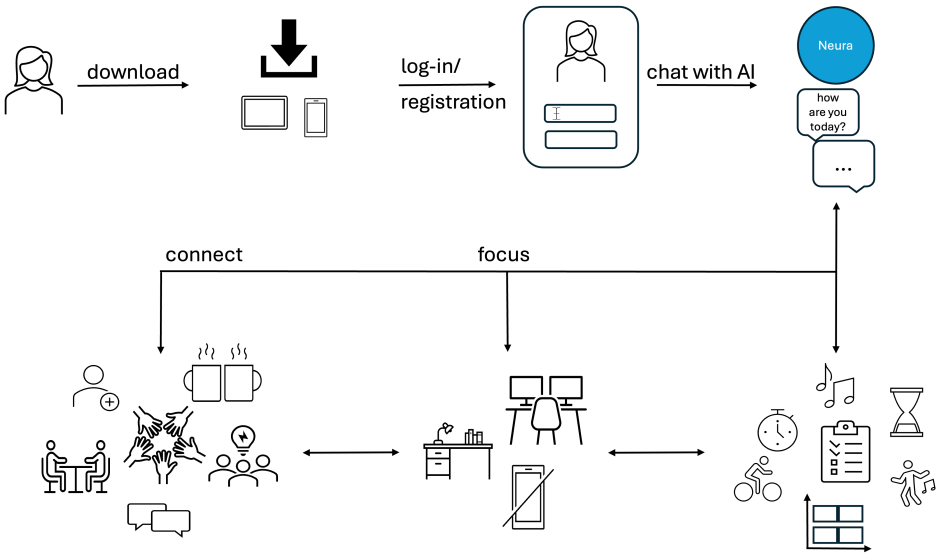


Figure 27: User-Flow Diagram

With the Focus functionality, the user sees an overview of the tasks to be completed. It is possible to have complex tasks broken down into subtasks. The user can select a task that they would like to work on next and thus switch to focus mode. Neura can offer support and techniques such as the Pomodoro technique, or the user can search for support techniques themselves using the Browse function. If the user cannot concentrate at the moment, Neura suggests alternatives to compensate for this. Going for a walk or listening to music can help to clear your head so that you can focus on work again afterwards.

If you feel the need to talk about your problems or simply want to meet new people, use the Connect function. Here you can write to people in a chat or arrange to meet up, either for a coffee or to be productive together. You can also share your activities to motivate each other or to get and offer support.

You can switch back and forth between the functionalities, depending on what the user needs at that particular moment.

6.4 Integration requirements

Table 4: Integration requirements

Component	Purpose
OpenAi GPT-4 API	The GPT model forms the basis for Neura, which is the personalised AI-Assistant using individual user data.
Firebase Cloud Messaging	Push notifications and reminders of upcoming tasks and to implement the function of 'remind me 1 hour before' a meeting.
Weaviate Vector Database	Store vectors
PostgreSQL Database	Storing relational data, such as created meetup informations, users tasks or account data.
Keycloak	as the identity and access manager for the app log-in.
Apple Calendar Integration	to access and manages user apple calendar using the EventKit Framework
Apple Location Services	provides users device location. used to implement the 'share location' function when joining or creating a meet-up.

6.5 Scalability consideration

Our app follows a modular architecture approach, allowing individual components to scale independently. The key aspects to consider for scalability are services like the notification service, but especially the database and the API. Firebase Cloud Messaging scales automatically and can deliver up to 600,000 messages per minute, ensuring reliable delivery even as user numbers grow. [2] The other cloud-hosted databases used also support automatic resource allocation, allowing the system to handle growing user demand efficiently. Google Cloud SQL monitors storage usage every 30 seconds and automatically increases capacity when needed, up to a maximum of 64TB. Once this limit is reached, additional databases should be added and operated in parallel to partition the data load and ensure continued scalability. [3] The most critical point is the API access to ChatGPT. There are API rate limits, but these are expected to automatically increase with higher usage, allowing automatic scaling to accommodate a growing number of users. The highest tier allows up to 10,000 requests per minute, which should be sufficient even at large scale. [4] Naturally, the scaling of the different components is associated with increasing costs that need to be managed carefully.
In conclusion, our app is designed to handle increasing user numbers and scale efficiently. This is crucial because one of the key features, like meetups, truly comes into its own with a large and active user base.

[2] Firebase Cloud Messaging Documentation, accessed July 2025
[3] Google Cloud SQL Documentation, accessed July 2025
[4] OpenAI GPT-4.1 API Documentation, accessed July 2025

7 Risk Management

In the following, we want to point out what could go wrong with our prototype, why users might not use it, and what outside factors could affect the project. We also explain how we plan to reduce these risks and how we make sure our solution is fair, safe, and works for different people.

The **technical risks** associated with our idea are primarily linked to the use of APIs. Specifically, if our app integrates with a GPT model from OpenAI, or any other model by a different firm, we are vulnerable to changes made by the provider. Such changes could significantly impact our app's effectiveness and user experience. Potential changes include limitations on API requests. This raises the question of user acceptance if, for instance, users could only interact with Neura five times a day. This would of course massively impair brain-dumping via Neura and the automatic creation of subtasks. Furthermore, increased costs are a risk, which could undermine the app's financial sustainability. This would force us to consider whether users would be willing to pay for ongoing access or if alternative revenue streams, like in-app advertising, would be required to offset the elevated expenses. A more extreme risk is the complete discontinuation of API access. In such a scenario, we would have to ascertain if an equivalent model could serve as a seamless 1:1 replacement, both in terms of quality and costs. Beyond these API-related concerns, it is also crucial to acknowledge that inaccurate AI results can lead to user frustration and even disorientation. If the AI outputs are not sufficient and do not meet users' expectations, it undermines user trust in NeuroConnect and can heavily impact usage. Another technical risk involves device compatibility. ADHD affects individuals across all demographics. An application that is memory-intensive due to AI functionalities or animations like animated bubbles could experience stuttering and crashes on older devices. This would considerably impair accessibility and user-friendliness for a part of our target group that does not own the latest smartphone generation.

Moving from technical considerations, we now turn our attention to **user adoption risks**. The first risk identified appears during the registration process. We need to find the sweet spot between enabling deep AI personalization and keeping the questionnaire concise. Our goal is to offer Neura as an AI assistant customized to individual user needs, ensuring maximum utility. However, the pitfall is that an overly long questionnaire can be off-putting, leading users to abandon the registration process out of frustration. Following on from this, we also face the risk of general scepticism towards an AI-based approach. Many people have already had experiences with an AI and some might be fascinated by its abilities and others might be dissatisfied with its output, perhaps due to poor models or even the problem of hallucinations. This disappointed subgroup of potential users could be critical of such an application. To address this, giving users the option to turn off the AI would be a quick solution. While that removes a central part of the app's core functionality, the app would still offer the informative browse feature and the community hub. In addition to scepticism, there is the risk that the app may become overwhelming due to an excessive number of features. A rich feature set may demand a lot of time and effort to explore and understand, which may become counterproductive. Moreover, a user might feel overwhelmed by the variety of features and quickly turn away from the app. An onboarding process that explains the features to newly registered users would help with this risk. The same risk also applies to offering too many customization options, which in turn can have an overwhelming effect and result in a blockage rather than helping. To counteract this, our initial colour scheme tries to offer a calm user experience, avoiding cognitive overstimulation.

Another critical aspect is tied to the community feature itself and the importance of **network effects**. Specifically, we are dealing with the classic 'chicken or egg' dilemma: we need users who want to join community meet-ups, but to attract them, we also need users who are willing to organise and host those events. Each group depends on the other, and for the community feature to be truly useful we need to encourage both types of users to engage with the app. Following on from this, the decline in motivation is another challenge. Not only can users turn away from the app due to the reasons just mentioned, but notification fatigue or boring gamification might also reduce long-term engagement. Thus, it will be particularly important app usage leads to positive results, increasing the likelihood of continued use.

When it comes to **external factors** that could significantly impact our app, we identified two main areas of risk. First, as we operate within Germany, our app must comply with German and EU regulations and directives. This includes current and upcoming legislation, such as the EU AI Act, which presents uncertainty and potential risks that are difficult to anticipate at this stage. While we cannot control the regulation imposed on us, we can ensure our team remains agile and adaptable, allowing us to respond effectively to any changes. But one thing we do have influence is to consider working with a European AI firm. For example, Mistral AI, as a French company, develops models that are subject to EU regulations, potentially minimizing compliance issues compared to using models from U.S.-based providers like OpenAI. However, the quality and suitability of Mistral AI's models would still need to be evaluated to ensure they meet the technical and functional requirements of our app. A second major potential risk is the potential loss or shortage of skilled personnel. Developing and maintaining a high-quality app experience requires access to talented personnel, which in turn demands adequate financial resources, in addition to the financial burden of server capacities and API costs. While monetization through in-app advertising is an option, as already briefly mentioned above under technical risks, it would negatively affect user experience. Instead, securing funding through investment rounds, would be a more suitable way to support the long-term product development. Giving up equity for financing is part of the start-up business and would be the most straightforward way to go.

Social and Ethical risks mainly relate to accessibility. It is essential that the use of the app is free of charge. Since equity financing is limited, sponsorships or partnerships with established and large ADHD organisations, such as Children and Adults with Attention-Deficit/Hyperactivity Disorder (CHADD) or the Attention Deficit Disorder Association (ADDA), could be a sustainable way to secure long-term free access for users. Such sponsorship would also raise the app's profile, attract more users and strengthen the value of the forum and community features through network effects. Beyond accessibility, cultural sensitivity should also be taken care of. Cultural differences between users should be taken into account within the browsing experience. The initially suggested

coping strategies will be tailored to each individual user. Similarly, cultural differences can also be considered when displaying recommended forum topics, by analysing anonymised user profiles and basing suggestions on similar profiles. To ensure the highest level of data protection, these profiles should be fully anonymized, for example by implementing k-anonymity, to prevent any security breaches that would lead to re-identification of individual user. Additionally, building a community platform also brings several risks related to user behaviour. Toxic behaviour or conflicts can arise, and shared content may be unqualified or misleading. The effort required to moderate such content can be time-consuming and requires expert knowledge.

8 Legal and Ethical Considerations

In the following, we want to explain what user data we collect and how we keep it safe. We mention if the app is made for vulnerable groups like children or the elderly. We also talk about making the app accessible for everyone, following legal rules. Then, we note any copyrights or trademarks that matter, and make sure the app follows platform rules.

The following data is collected by the app: personally identifiable data such as name, email, password, place of residence (needed for local meet-ups) and a profile picture. Further, personal ADHD data is retrieved through the questionnaire and usage data such as the created and joined events, focus tasks, and user behaviour (to analyse productivity), is collected. These represent the minimal and necessary data needed to ensure a meaningful app usage and is only used for this purpose. For reasons of transparency, the user is informed about the collection and use of their data during the registration process. Data is stored on secure servers in Germany. To ensure confidentiality, encryption is used for both transmission and storage. Moreover, we also apply k-anonymity to protect against identification of individuals from processed or aggregated data.

Our app serves a group that faces cognitive challenges and potentially emotional instability. This is the very reason why it is important that forums must have strict moderation to protect users from inappropriate interactions and triggering content. Certain topic forums may be hidden from minors to prevent exposure to content that could be inappropriate or emotionally distressing. It should be noted here that there needs to be a meaningful way of verifying user age. A mere question at the beginning would be pointless and would undermine protective mechanisms for vulnerable users, such as minors in this case. Further, children under a certain age should be asked for parental consent, possibly by email link confirmation, if they want to use the forums or be part of certain community meet-ups. This is for their safety.

9 Social Impact Assessment

NeuroConnect aims to create a long-term and meaningful impact for individuals with ADHD by addressing focus challenges but also emotional well-being, self-understanding, and community support. The app is designed to meet users where they are, helping them to better understand and cope with their neurodivergence - rather than working against it.

We would measure success and failure through a combination of quantitative and qualitative metrics. On the quantitative side, we would track metrics such as the number and duration of focus sessions, activity in the community forums, use of the AI assistant Neura, and the number of real-life meetups planned or joined via the app. Qualitatively, we would assess improvements in users' perceived ability to start tasks, emotional state, sense of belonging and self-efficacy. We would collect those insides through optional check-in prompts, surveys, and more user interviews. We would view any improvement in user productivity and well-being as success, whereas we would view it as a failure if and only if our app hindered our users from any improvement or if no form of social interaction would take place.

To capture user transformation narratives, NeuroConnect could have reflective features such as journaling and progress tracking, maybe even with data analytics. We would encourage users to share their experiences through anonymous stories, maybe expanding our forum topics with another section specifically for this progress. In terms of behavioural changes, we would love to see increased ability to start and finish tasks, easier access to peer support, and a reduction in shame and stigma related to ADHD.

Our vision for NeuroConnect would be to remain impactful even without our team's direct involvement. For this, we would engage with ADHD coaches, therapists, and pedagogues to deepen our app's impact. Sustainable funding could come from a freemium model or public-sector innovation funding, whereas we would prefer the latter as we believe that such an app and all its functions should be available for everyone. To ensure a safe space within the community, we would equip all connectivity features with moderation tools and guidelines for safe discussions and collaborations.

Long-term user engagement could be supported by adaptive content, evolving features based on community feedback, and gamification elements or social incentives such as shared progress and accountability. As our app is community based, we think that once our users find a strong community of like-minded people and really use the app as a tool for productivity (especially through joining productivity meet-ups) it would automatically result in longer user engagement. Further, we envision that happy users recommend the app to other neurodiverse friends or people they know, so we could naturally let the app's community grow.

If the app were to stop working tomorrow, many users would lose a system of structure, support, and connection that they may not easily find somewhere else. This highlights the importance of building a solution that is sustainable, community-driven, and resilient.

11 Conclusions and Key Learnings

In the following we want to give a short summary of our application and the key learning that we as a team had during the project.

Starting with the idea of creating an app that helps people with ADHD to be productive and manage their daily lives, to another idea of setting aside the productivity aspect and instead creating a kind of social network for people with ADHD so that they can interact with other people who understand these problems, our NeuroConnect app was born. It contains a mixture of the aforementioned approaches. Both the support for everyday life and productivity as well as the possibility to get in touch with other people found their way into our final product. There is also another component that enables users to learn about methods that can make life with ADHD easier.

We started the project with an open mind. None of the team members had any previous experience with the design thinking process. With the knowledge that we didn't have to implement anything, but could simply live out our own creativity, we started work in a positive frame of mind. The first interviews were quickly conducted and the initial requirements for the app crystallised. At the same time, it quickly became clear that every user is individual and that the expectations of an app like the one we had in mind can also be very different. So you had to decide what and how to implement things, as you can't realise every wish and certainly can't satisfy every potential user completely. In addition to the requirements and ideas from the interviews, we also came up with our own ideas. All in all, this meant that we often had the problem of not knowing where to start, which design or which functionality to choose. This resulted in a major key learning: it is almost impossible to develop a perfect product with the first attempt and design thinking is an iterative process. You first have to decide on a variant in order to make progress and then you can get feedback and may have to change or delete things that you thought were good and sensible and completely rethink them. Which brings us to the next challenge: feedback. The initial interviews went perfectly. The interviewees were creative and described their needs and concerns. However, when it came to looking at a prototype, imagining the functionality and evaluating it, the willingness to do so decreased. This involved too much effort for the interviewees. We were able to counteract this somewhat by asking very precise questions and explaining the functionality in detail in order to give people as clear a picture of the product as possible. The final challenge was the design thinking process itself. The freedom within the process and the iterative working method were both a blessing and a curse. Not only the feeling of losing the overview, as described above, but also the occasional lack of structure was a challenge for us. We met this challenge with enough patience and lots of questions for Isabella.

In addition to the challenges, which have contributed significantly to each team member developing and making progress together, we have gained experience in this project. This is primarily due to the fact that we used the design thinking process ourselves for the first time, but also due to all the other things that were necessary to finalise the project. This includes, for example, creating a mock-up in Figma, writing a report in English or working in a team (see chapter 13), but also things that we are not yet aware of, that we cannot yet put into words, but that will help us in the further course of our careers.

In general, we can say that we complemented each other well and worked very harmoniously as a team. The only exception to this was the attempt to finalise texts, change the prototype or fill in other documents during Zoom meetings in which all four team members were present. The problem here was that nobody really felt responsible and at the same time there were far too many topics in the room at once or being discussed in parallel and therefore nothing got done. However, we quickly realised this and continued to divide up such tasks and then work on them in small groups of two if necessary.

In conclusion, it can be said that we worked very well as a team and complemented each other very well with our different strengths and weaknesses, which made for a perfect project in terms of time spent together.

12 Poster and Video

Poster:

Neuro-Connect
An Organising Support for Neurodiverse Minds

A Computing for Social Good Project – Summer Semester 2025

TECHNISCHE UNIVERSITÄT DARMSTADT

Why this App Matters

- ADHD affects over **5% of adults globally**
- Many struggle with **focus, time blindness, procrastination**, and **emotional overload**
- Existing tools (e.g., calendars, to-do lists) tools don't work for ADHD, they are often…
  - **Too rigid**
  - **Too complex**
  - **Not ADHD-aware**
- ADHD is not just about attention, it's about executive functioning: the brain's ability to plan, prioritize, and stay focused.

"I always have 10 tabs open, in my head."

"I want to be productive, but I don't know where to start."

"If I don't capture my thoughts right away, they're gone."

Importantly we don't want to "fix" **ADHD**, we want to **support** it.

Impact at a Glance

In conclusion, our results matter and pave the way for future developments in this area. Key outcomes include:

- More than **7 ADHD interviews** guided our features
- **3 rounds** of design and feedback iterations
- Identified **3 key challenge** areas: Time Management, Organisation, Mental Load
- Built a **Minimal Lovable Product**, not just an MVP
- **100% of testers loved** the app by the final iteration

Our Most Loved Feature was the Brain Dump Functionality

NeuroConnect creates social good by providing...

Structure – without strict rules

Learning – through stories and shared methods

Consistency – via micro-habits and nudges

Guiding our Project was the Desing Thinking Framework



Empathise **Define** **Ideate** **Prototype** **Test**

What NeuroConnect Offers

1. **Brain Dump**
   Capture unstructured thoughts via text or voice. AI turns them into actionable tasks or goals.
2. **Focus Mode**
   Visualise and track tasks with smart productivity bubbles. Create focus sessions that adapt to your energy levels.
3. **Connect**
   Build real-life and digital communities for ADHD peer support. Celebrate goals, join co-working meetups, stay motivated.
4. **Browse**
   Curated strategies for focus, planning, and self-care. Read tips, share experiences, learn what works, from real people.



We designed **NeuroConnect** as a smart, supportive app that works **with** neurodiverse minds, not against them.

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Watch the Video

Test it Yourself

14 Disclaimer

In this section we want to shortly describe how we used AI tools.

We jointly agreed to limit the use of AI to the necessary minimum in order to gain the most from the Design Thinking method, as letting all tasks be done by AI would prevent us from being fully immersed in the project and could also keep us from fully understanding the problem we have identified, as well as why and how we are solving it. We use AI to improve our written text, namely grammar, spelling and, if necessary, to translate German text into English. We also use AI to speed up the transcription of interviews, followed by manual clean-up and correction and lastly to generate the sketches of our user personas. In general, for each task where we employ an AI tool, a team member provides the original work as input, equally the output is reviewed by a team member. The result is corrected and used as a basis for further steps.

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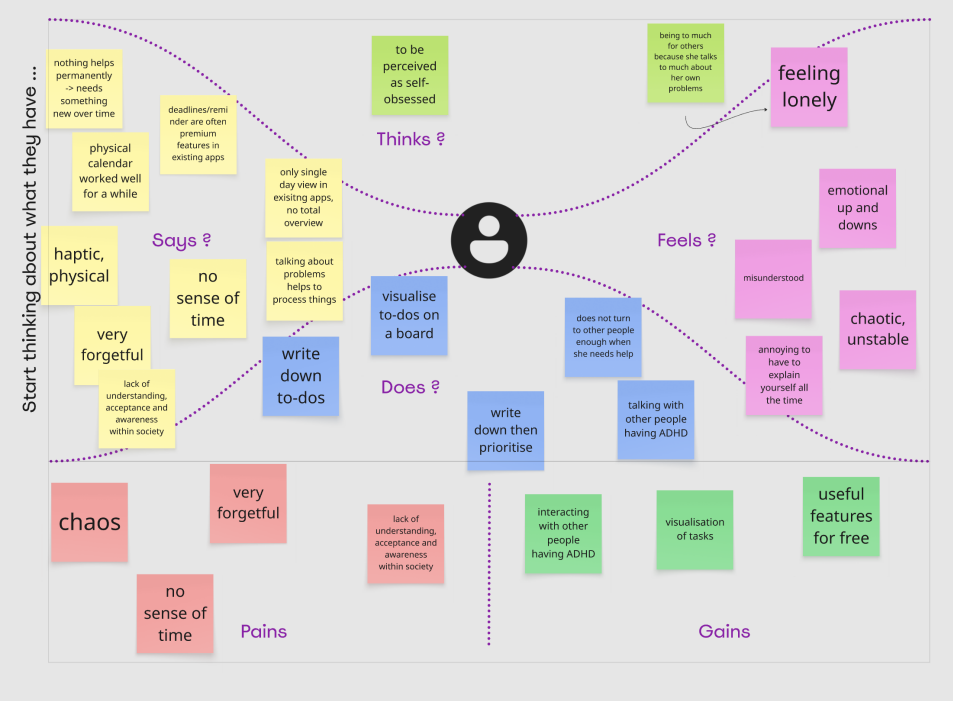


Figure: Empathy Map Ines

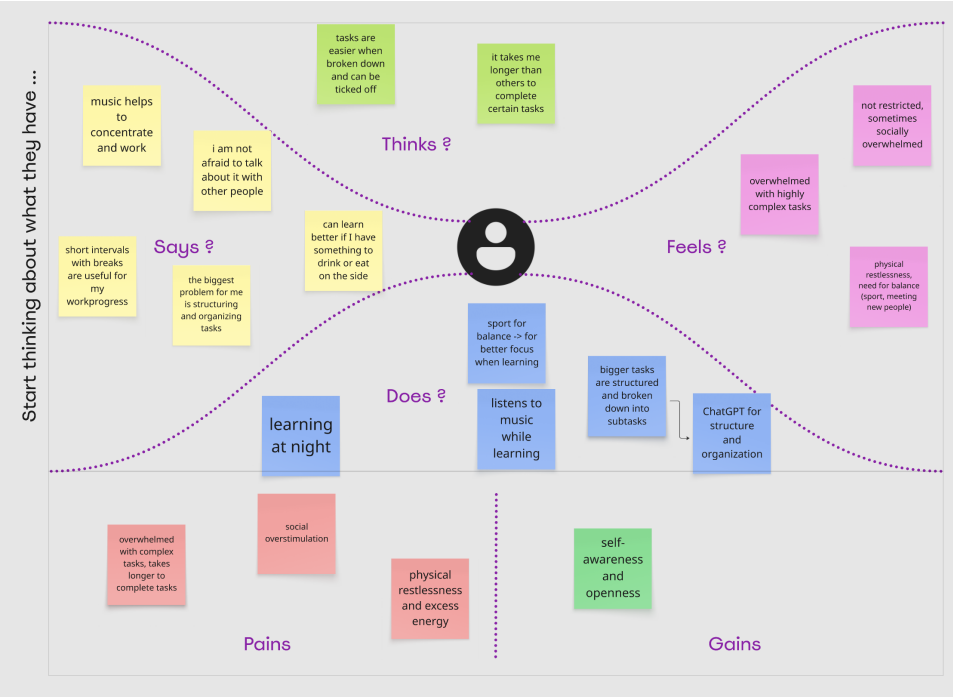


Figure: Empathy Map Stefan

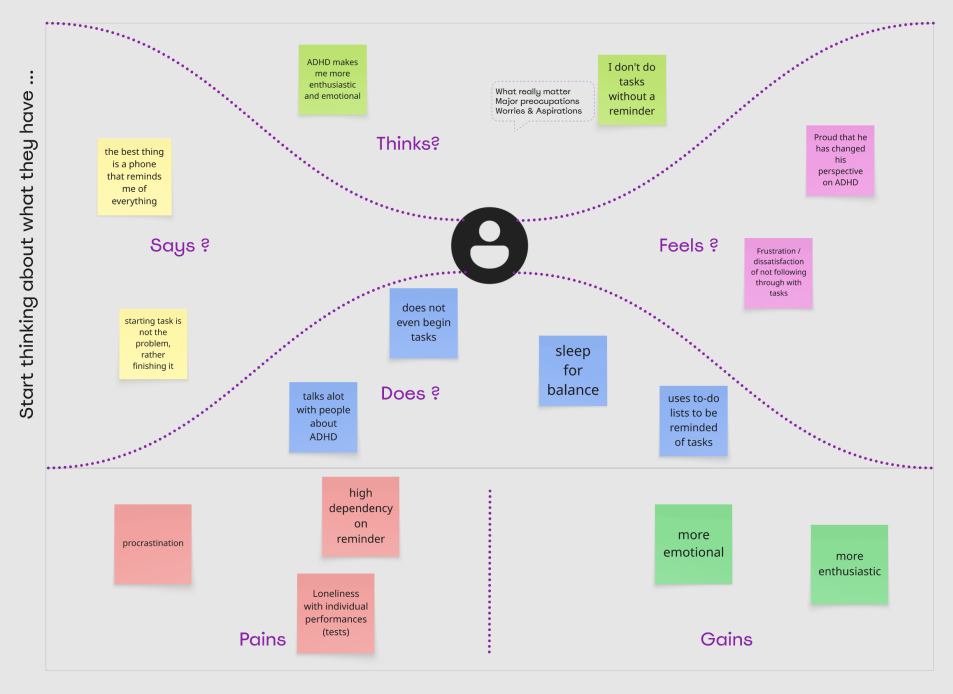


Figure: Empathy Map Leon

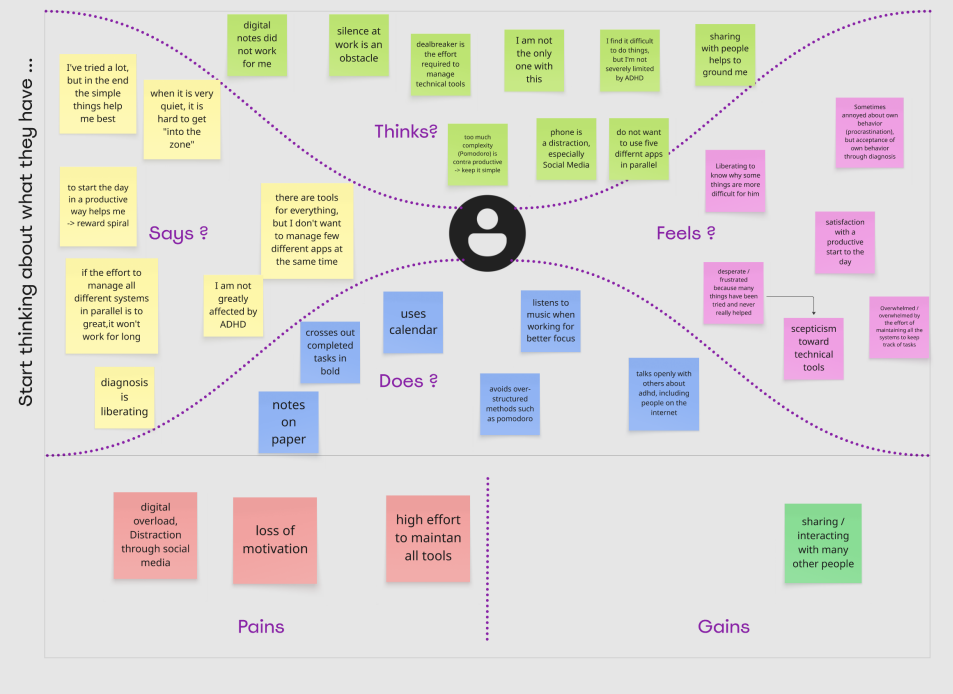


Figure: Empathy Map Dennis

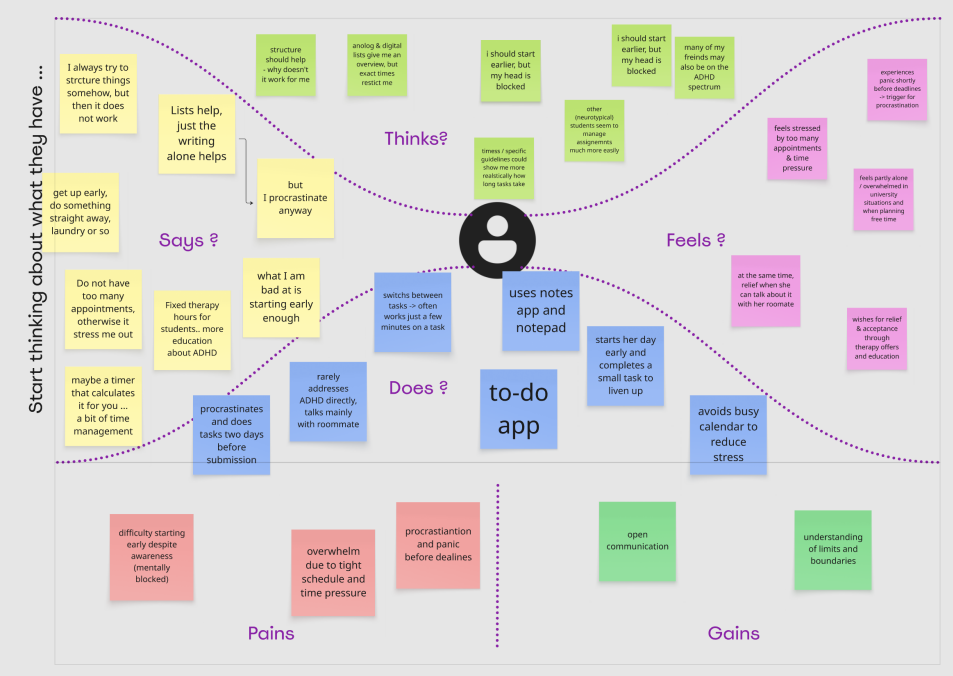


Figure: Empathy Map Franzi



Figure: Empathy Map Lala

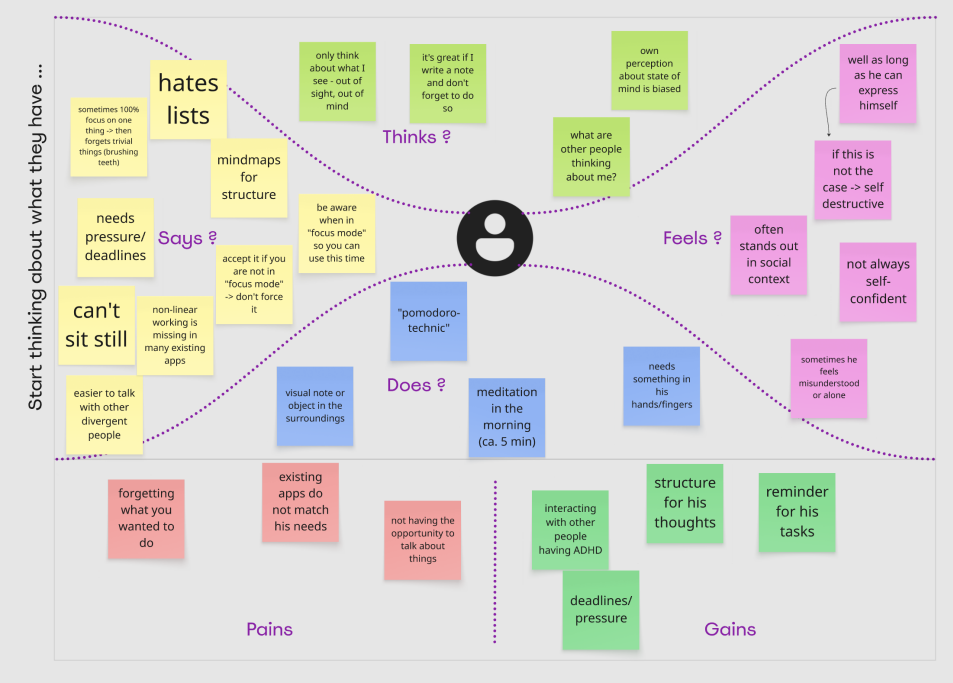


Figure: Empathy Map Ray

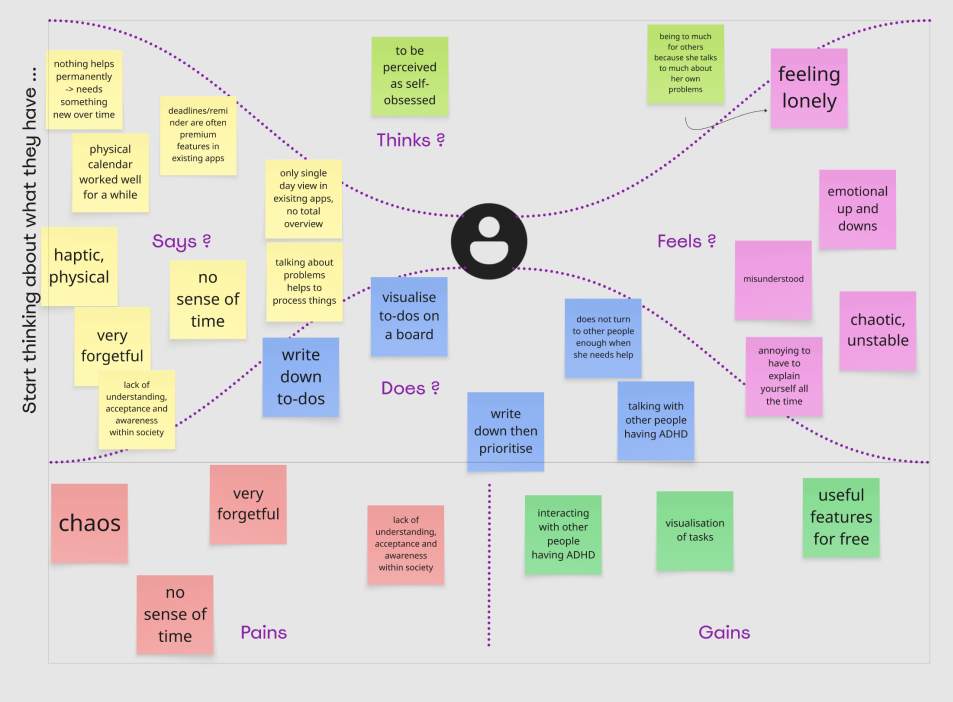


Figure: Empathy Map Ines